

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
            #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)

```

```
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df

7952

Nombre de lignes df clean	7952
Nombre de features	29
% de NaN	0.0
Taille dataset entraînement	5566
Taille dataset test	2385
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_6h_mean, r_NH3_st_6h_std, r_NH3_st_6h_rms, r_NH3_st_6h_median, r_NH3_st_6h_min, r_NH3_st_6h_max, r_NH3_st_6h_diff_last, r_NH3_st_6h_slope, r_temperature_st_6h_mean, r_temperature_st_6h_std, r_temperature_st_6h_rms, r_temperature_st_6h_median, r_temperature_st_6h_min, r_temperature_st_6h_max, r_temperature_st_6h_diff_last, r_temperature_st_6h_slope, r_umidity_st_6h_mean, r_umidity_st_6h_std, r_umidity_st_6h_rms, r_umidity_st_6h_median, r_umidity_st_6h_min, r_umidity_st_6h_max, r_umidity_st_6h_diff_last, r_umidity_st_6h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

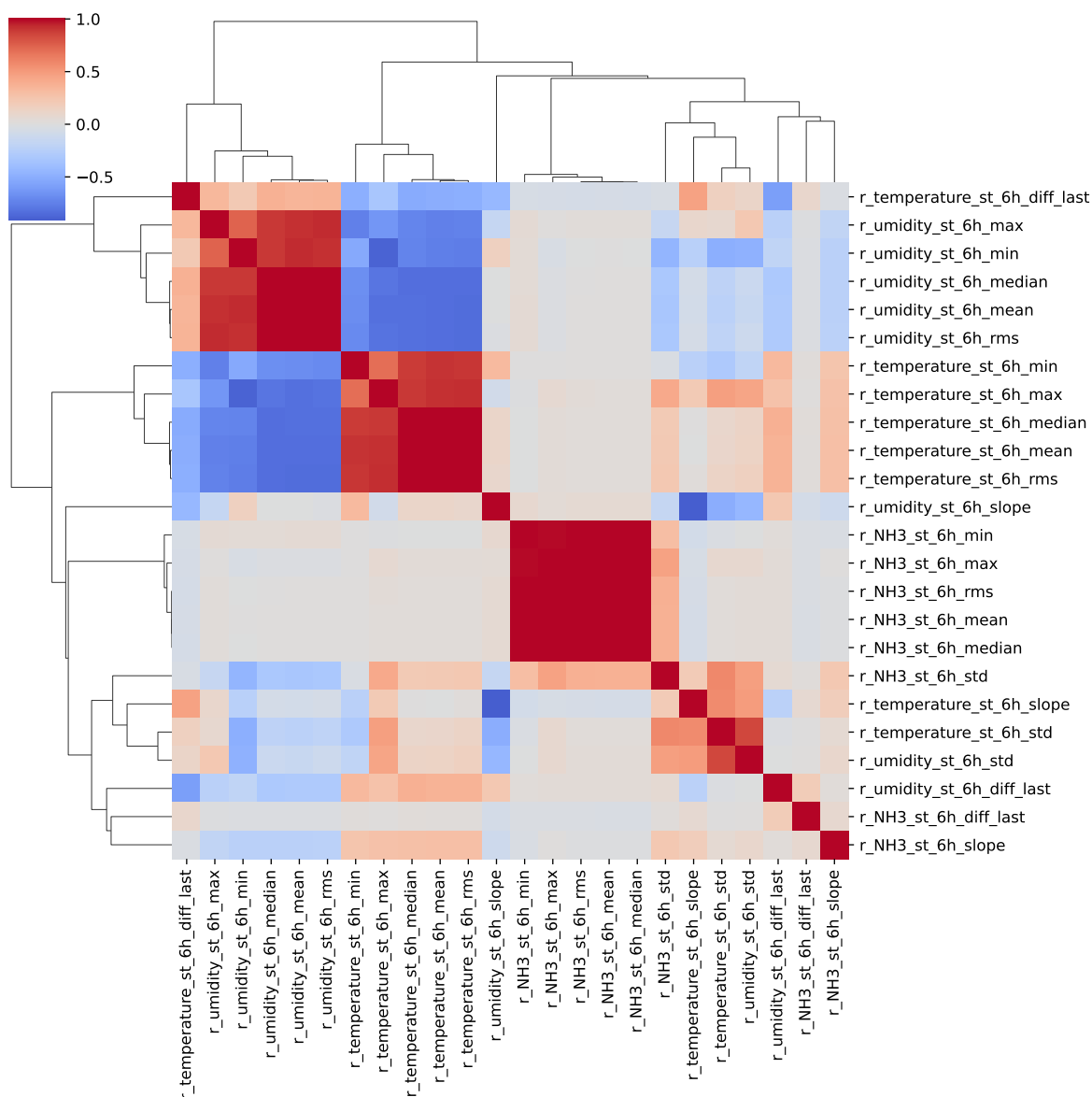
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_st_6h_max, r_NH3_st_6h_mean, r_NH3_st_6h_rms, r_NH3_st_6h_min, r_NH3_st_6h_median ; **cluster 7:** r_NH3_st_6h_std ; **cluster 8:** r_NH3_st_6h_diff_last ; **cluster 9:** r_NH3_st_6h_slope ; **cluster 10:** r_temperature_st_6h_max, r_umidity_st_6h_median, r_umidity_st_6h_min, r_umidity_st_6h_max, r_temperature_st_6h_rms, r_temperature_st_6h_min, r_temperature_st_6h_median, r_umidity_st_6h_rms, r_umidity_st_6h_mean, r_temperature_st_6h_mean ; **cluster 11:** r_umidity_st_6h_std, r_temperature_st_6h_std ; **cluster 12:** r_temperature_st_6h_diff_last ; **cluster**

13: r_umidity_st_6h_slope, r_temperature_st_6h_slope ; **cluster 14:** r_umidity_st_6h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_6h_max, r_NH3_st_6h_std, r_NH3_st_6h_diff_last, r_NH3_st_6h_slope, r_temperature_st_6h_max, r_umidity_st_6h_std, r_temperature_st_6h_diff_last, r_umidity_st_6h_slope, r_umidity_st_6h_diff_last



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_6h_max, r_NH3_st_6h_std, r_NH3_st_6h_diff_last, r_NH3_st_6h_slope, r_tempera-

ture_st_6h_max, r_umidity_st_6h_std, r_temperature_st_6h_diff_last, r_umidity_st_6h_slope, r_umidity_st_6h_diff_last

Len(Features_X): 14

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

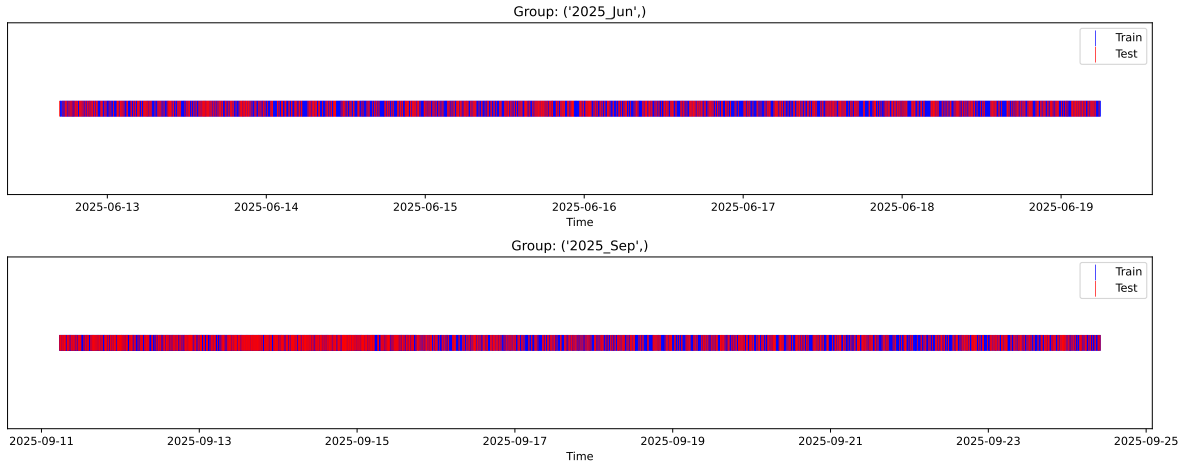
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5566 samples Test: 2386 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3896 samples Test: 1670 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3896 samples Test: 1670 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3896 samples Test: 1670 samples Excluded: 0 samples Window exclusion: 0h



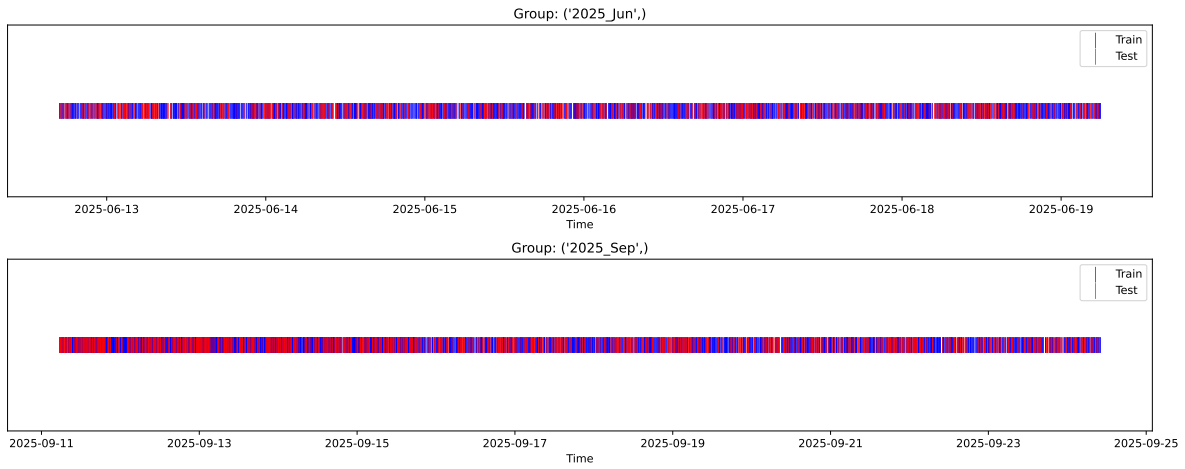
Fold 4/5

Train: 3896 samples Test: 1670 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3896 samples Test: 1670 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.349, std=0.796, min=0.959, max=13.845
 y_{test} : mean=2.366, std=0.808, min=1.011, max=7.822

Fold 0 — y_{test} : mean=2.39, std=0.82, y_{train} : mean=2.33, std=0.79

Fold 1 — y_{test} : mean=2.35, std=0.77, y_{train} : mean=2.35, std=0.81

Fold 2 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.35, std=0.80

Fold 3 — y_test: mean=2.32, std=0.75, y_train: mean=2.36, std=0.82

Fold 4 — y_test: mean=2.36, std=0.83, y_train: mean=2.34, std=0.78

train scores CV: [0.51387209 0.50112558 0.50581126 0.49808137 0.51545427]

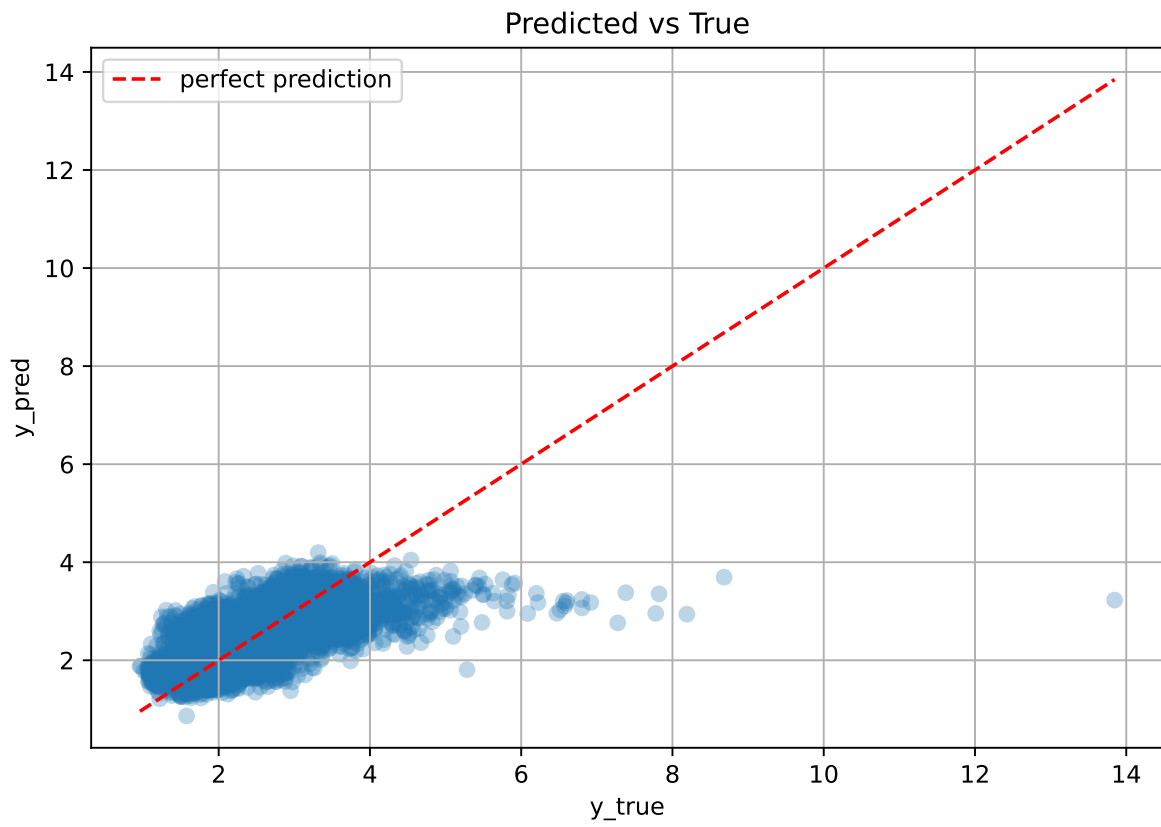
test scores CV: [0.47408752 0.50343828 0.49285379 0.50741483 0.47426938]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5069	0.4904	0.0141	0.4821	0.4833	-0.0072	0.9854	1.0336

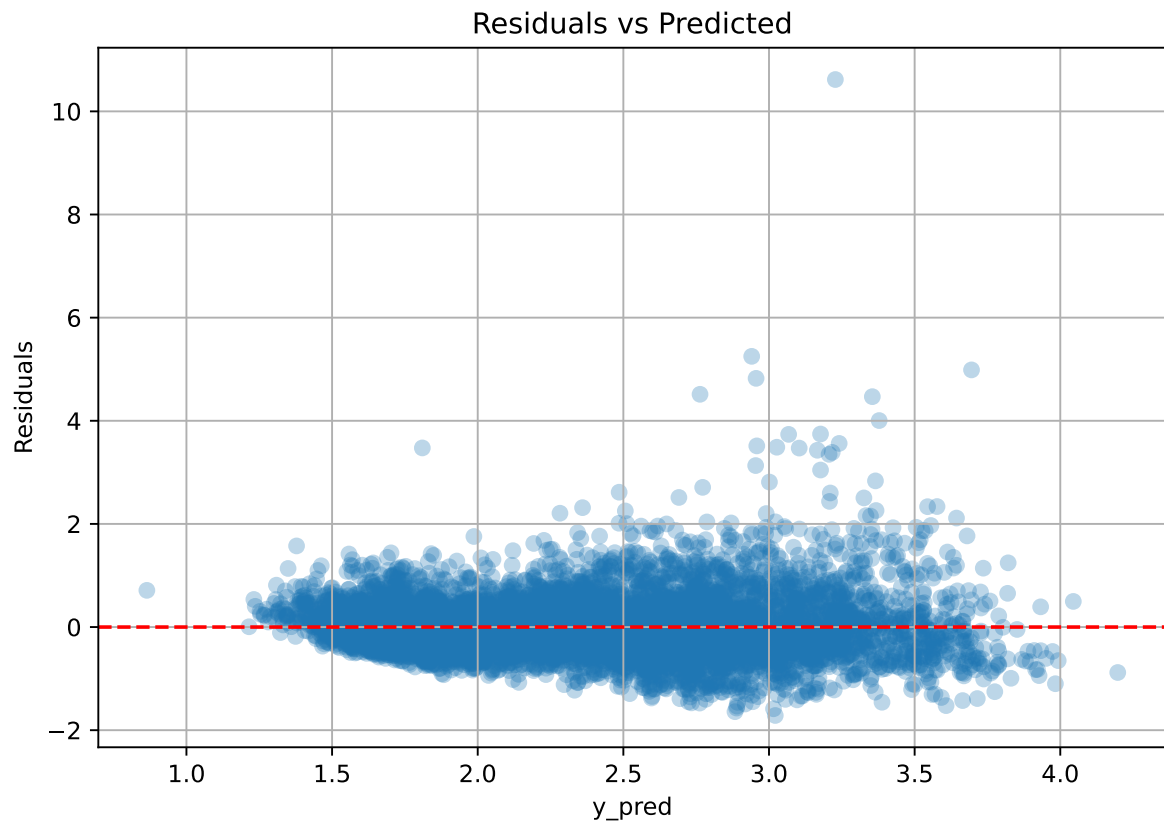
Coefficients:

coef_r_CO2 = -0.0628, coef_r_NH3 = -0.4268, coef_r_NH3_st_6h_diff_last = 0.0360, coef_r_NH3_st_6h_max = 0.3660, coef_r_NH3_st_6h_slope = 0.1009, coef_r_NH3_st_6h_std = 0.0269, coef_r_THI = 0.9359, coef_r_temperature = -1.0977, coef_r_temperature_st_6h_diff_last = 0.1899, coef_r_temperature_st_6h_max = -0.0266, coef_r_umidity = -0.6209, coef_r_umidity_st_6h_diff_last = -0.0922, coef_r_umidity_st_6h_slope = -0.0936, coef_r_umidity_st_6h_std = -0.0181, intercept = 2.3489,

3.4.2 Scatter



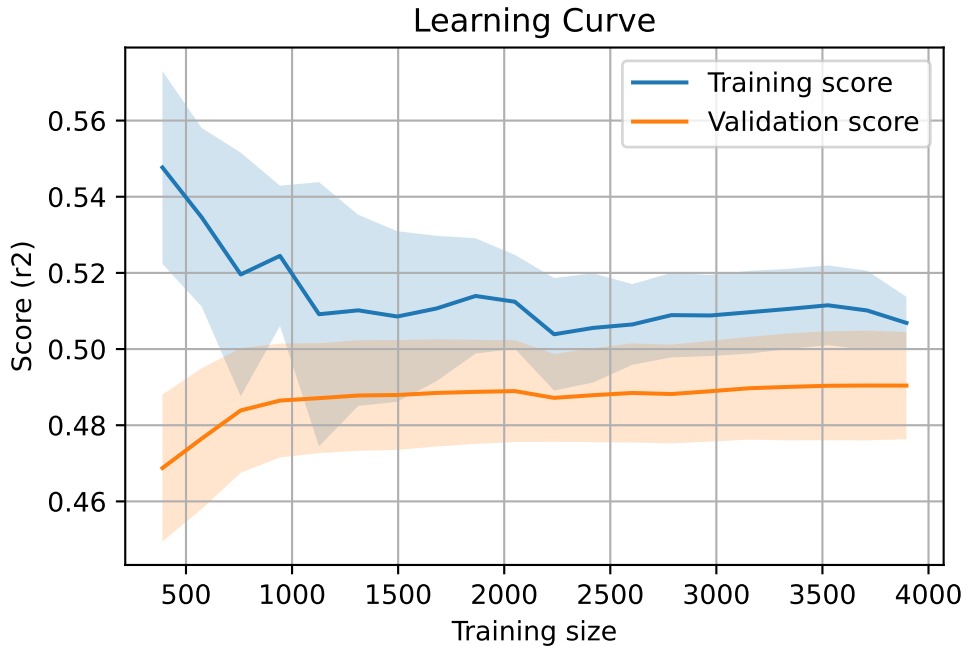
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7952, `Len(training)` : 5567,

`Len(training_real)` : 3897, `Len(test_of_training)` : 1670, `Len(test)` : 2385



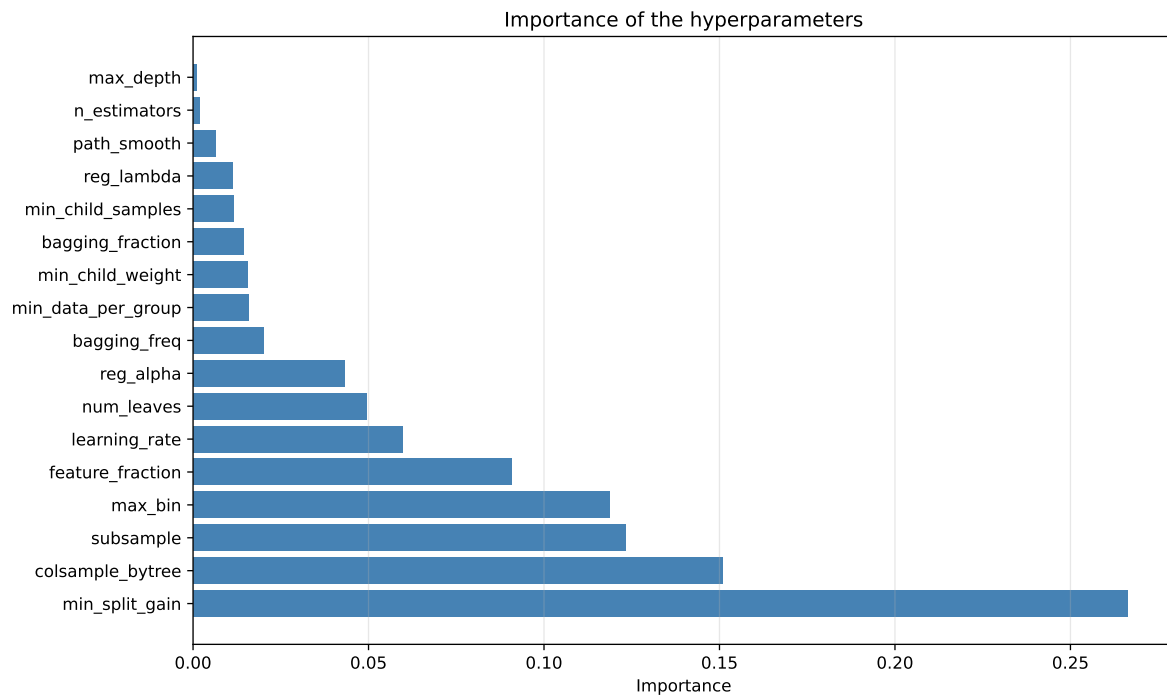
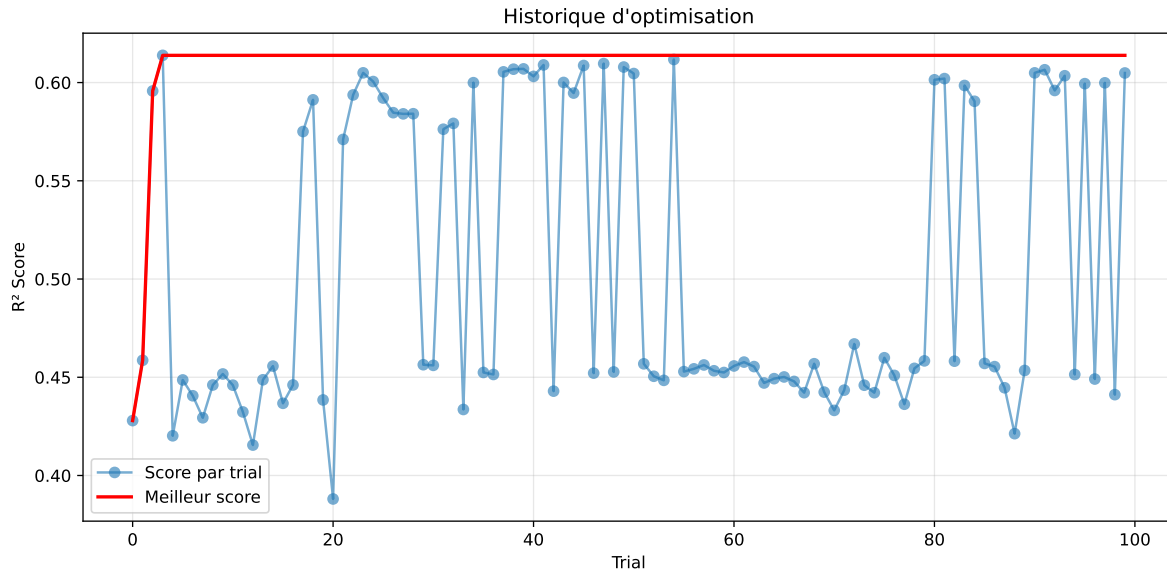
3.5 Model : LGBM

3.5.1 Gridsearch

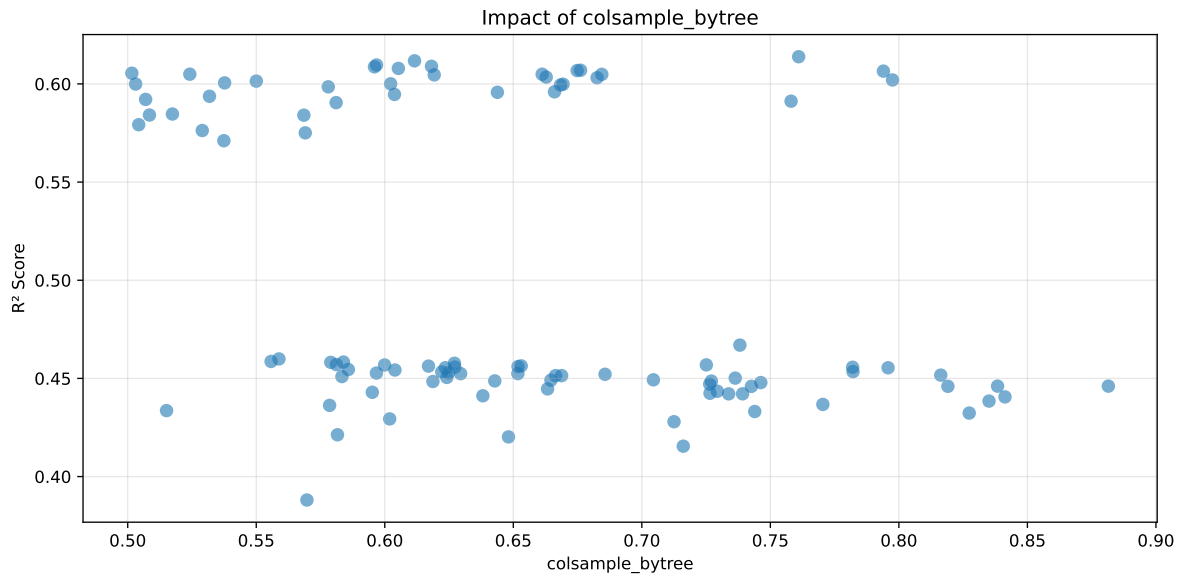
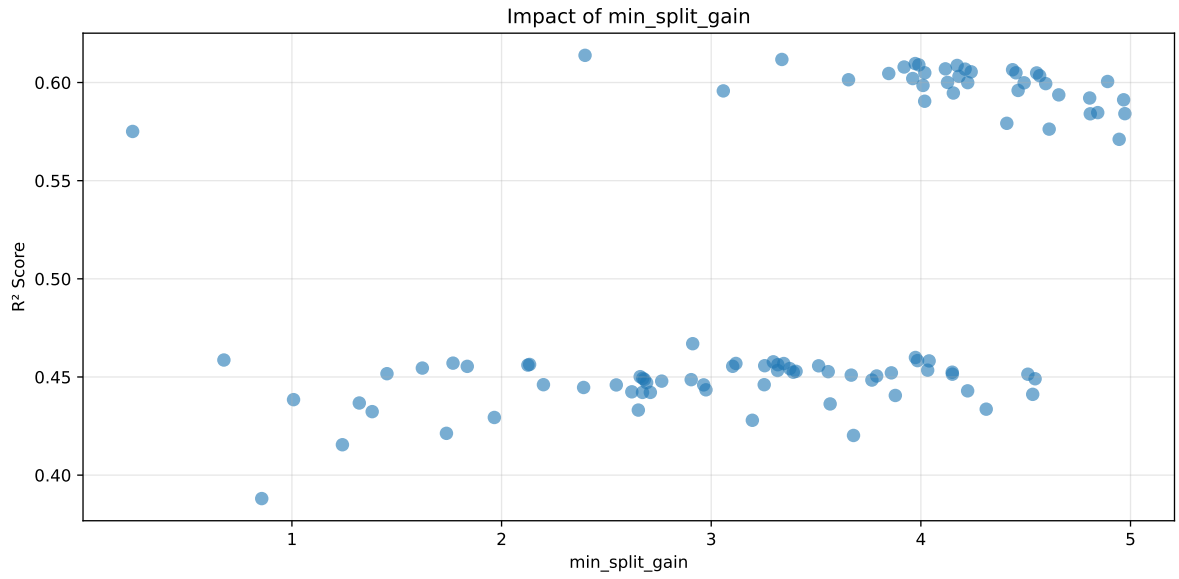
Distribution y_{train} vs y_{test} : y_{train} : mean=2.349, std=0.796, min=0.959, max=13.845
 y_{test} : mean=2.366, std=0.808, min=1.011, max=7.822

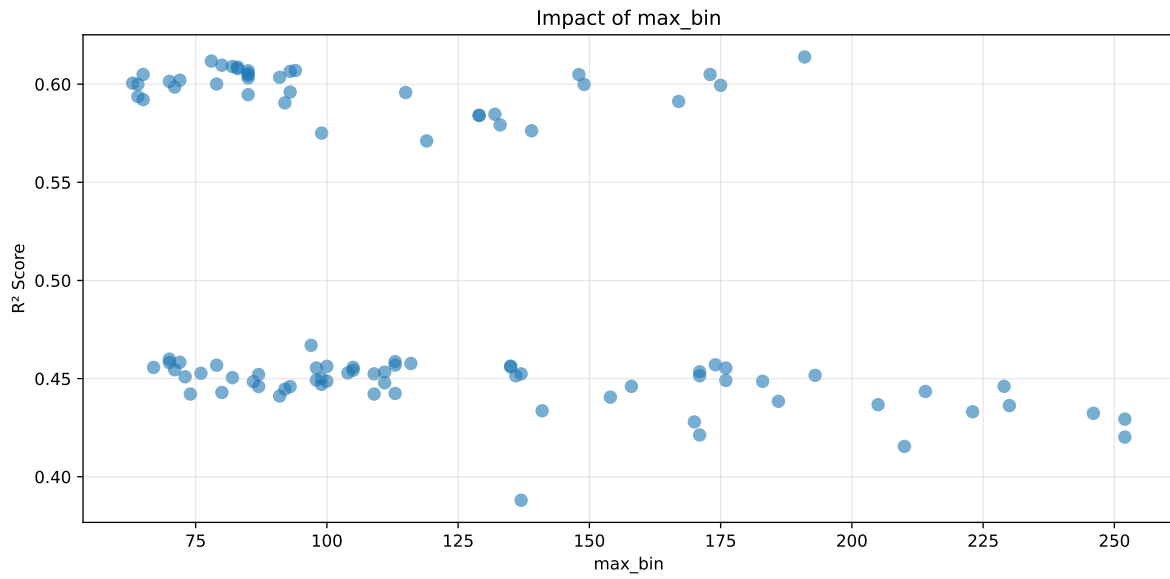
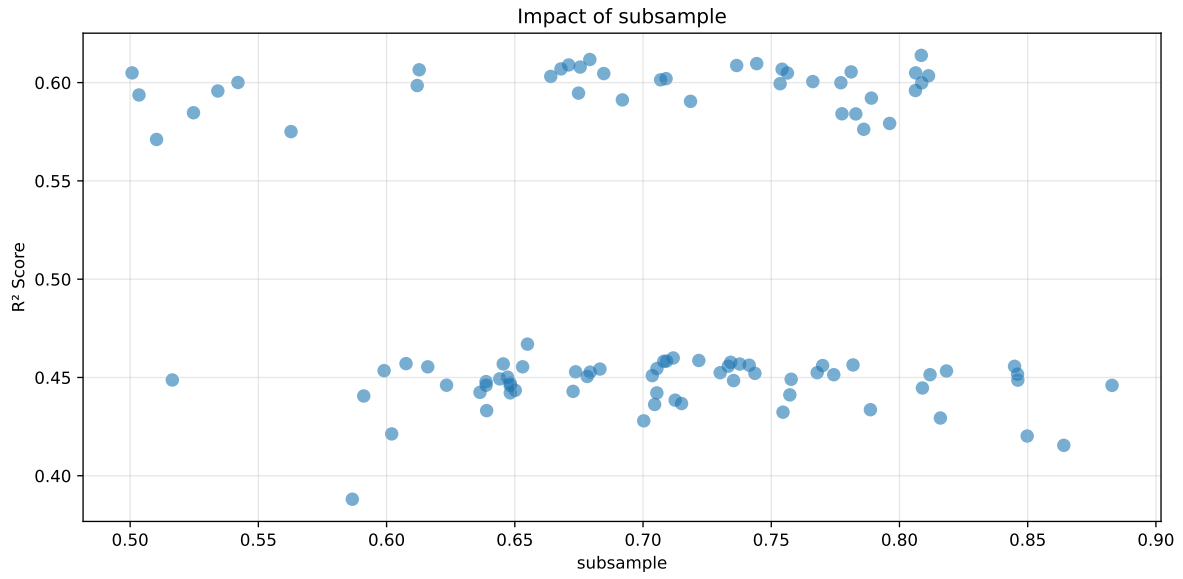
===== TOP Importance FEATURES =====

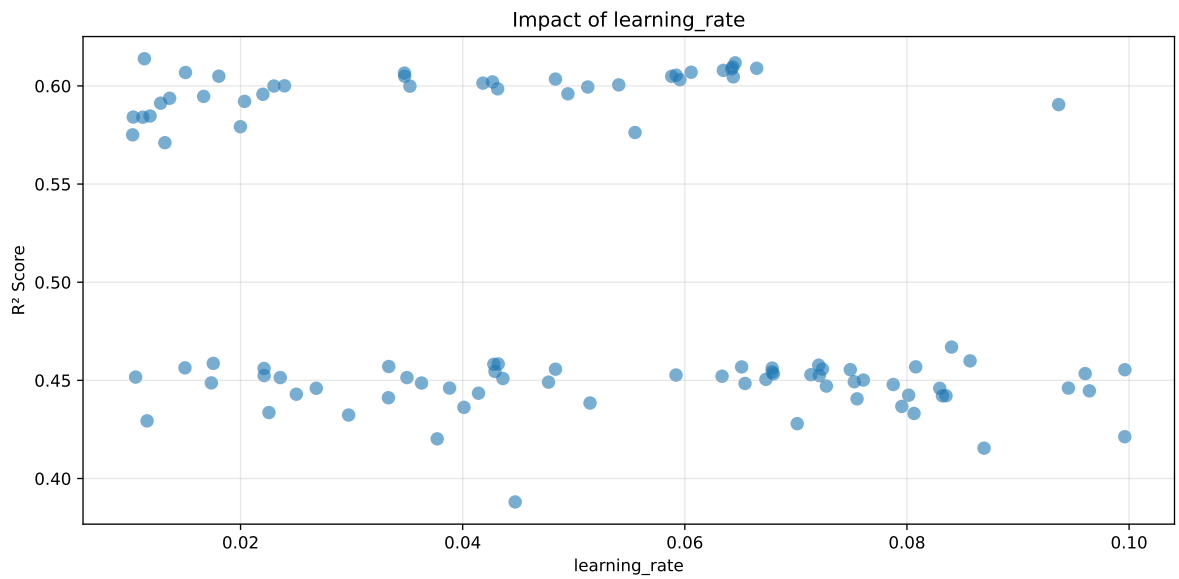
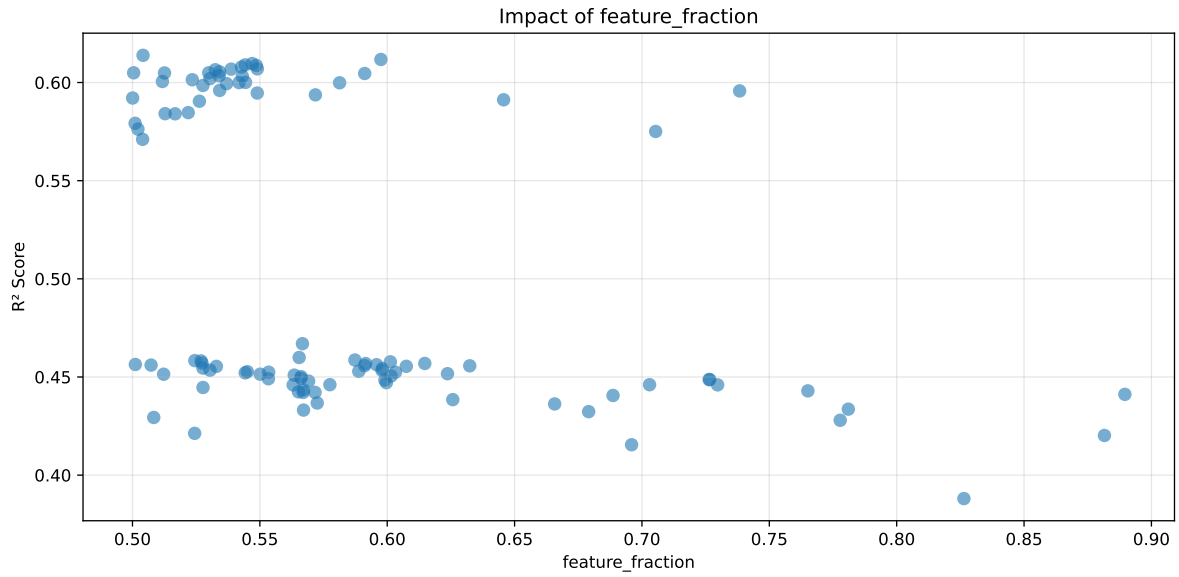
feature	importance
r_CO2	453
r_NH3_st_6h_max	309
r_umidity	285
r_temperature_st_6h_diff_last	263
r_temperature_st_6h_max	225
r_temperature	199
r_NH3	163
r_umidity_st_6h_diff_last	151
r_NH3_st_6h_std	143
r_umidity_st_6h_slope	140
r_NH3_st_6h_slope	131
r_THI	122
r_umidity_st_6h_std	103
r_NH3_st_6h_diff_last	92



Paramètres avec >5% d'importance : ['min_split_gain', 'colsample_bytree', 'subsample', 'max_bin', 'feature_fraction', 'learning_rate']

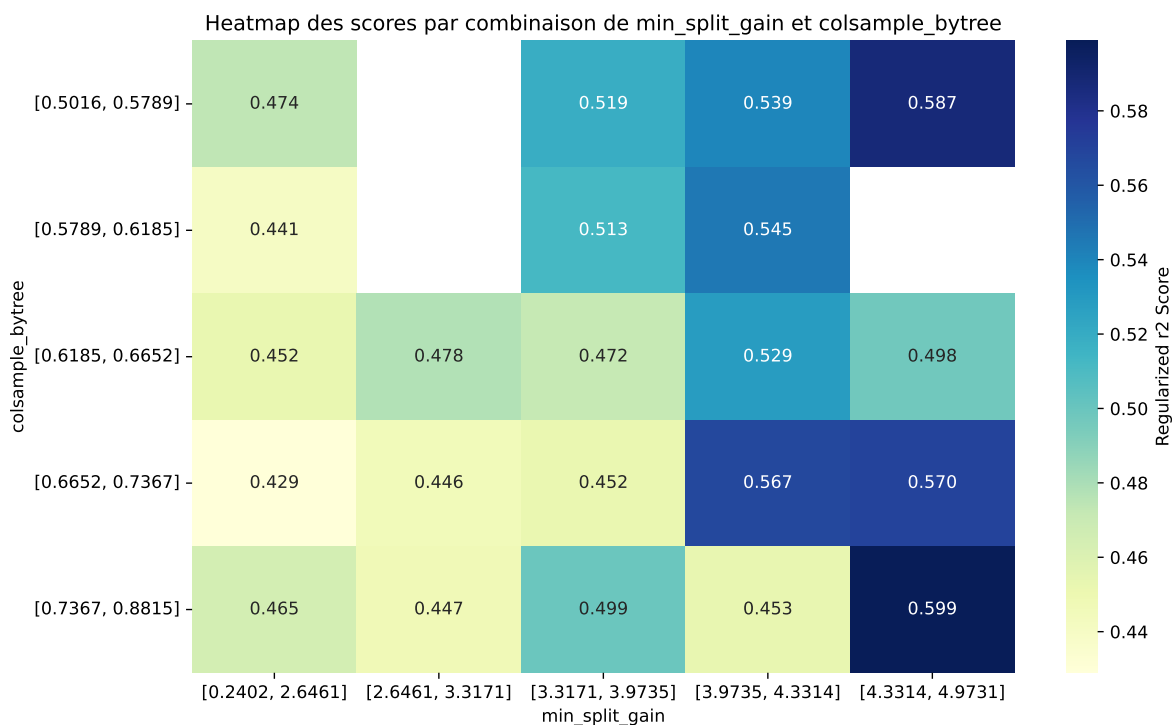






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6444	0.6138	0.6056	0.0224	0.6138	0.6275	0.0136	1.0222	1.0497
2	0.6422	0.6117	0.6019	0.0238	0.6117	0.6276	0.0159	1.0259	1.0498
3	0.64	0.6097	0.5994	0.0254	0.6097	0.6229	0.0132	1.0217	1.0497
4	0.6393	0.609	0.5986	0.0243	0.609	0.6189	0.0099	1.0162	1.0499
5	0.6387	0.6087	0.5986	0.0234	0.6087	0.6199	0.0112	1.0184	1.0493
6	0.6373	0.6079	0.5971	0.0237	0.6079	0.6215	0.0136	1.0223	1.0483
7	0.6364	0.607	0.5955	0.0247	0.607	0.6206	0.0137	1.0225	1.0484
8	0.6371	0.6068	0.5976	0.0257	0.6068	0.6238	0.017	1.028	1.0499
9	0.6365	0.6065	0.5977	0.025	0.6065	0.621	0.0145	1.0239	1.0495
10	0.6356	0.6054	0.5961	0.0248	0.6054	0.6219	0.0165	1.0272	1.0499
---	---	---	---	---	---	---	---	---	---
100	0.7277	0.6711	0.6697	0.0318	0.3881	0.6958	0.0247	1.0369	1.0844

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.011342453926835177, 'reg__num_leaves': 50, 'reg__subsample': 0.8085006969226423, 'reg__colsample_bytree': 0.7610142744148498, 'reg__min_child_samples': 28, 'reg__reg_alpha': 8.128788526642152, 'reg__reg_lambda': 8.259456459255205, 'reg__min_split_gain': 2.3981696590116575, 'reg__path_smooth': 0.7192006724699285, 'reg__min_child_weight':

0.008886981459978871, 'reg__feature_fraction': 0.5041401351860132, 'reg__bagging_fraction': 0.5933337982559577, 'reg__bagging_freq': 3, 'reg__max_bin': 191, 'reg__min_data_per_group': 121}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.06452965963669685, 'reg__num_leaves': 52, 'reg__subsample': 0.679245622673151, 'reg__colsample_bytree': 0.6115784586989641, 'reg__min_child_samples': 18, 'reg__reg_alpha': 2.6596629217305896, 'reg__reg_lambda': 44.472808085827786, 'reg__min_split_gain': 3.3369889105877393, 'reg__path_smooth': 1.115911797378437, 'reg__min_child_weight': 0.005023667657960254, 'reg__feature_fraction': 0.5975271279151864, 'reg__bagging_fraction': 0.7356475393218583, 'reg__bagging_freq': 2, 'reg__max_bin': 78, 'reg__min_data_per_group': 118}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.06430934751724872, 'reg__num_leaves': 58, 'reg__subsample': 0.7443254470982552, 'reg__colsample_bytree': 0.5968062753204119, 'reg__min_child_samples': 20, 'reg__reg_alpha': 3.556706706172256, 'reg__reg_lambda': 44.07329159244287, 'reg__min_split_gain': 3.973078202103488, 'reg__path_smooth': 1.9301990399944082, 'reg__min_child_weight': 0.00344599130192005, 'reg__feature_fraction': 0.5470070220962635, 'reg__bagging_fraction': 0.896700402568514, 'reg__bagging_freq': 3, 'reg__max_bin': 80, 'reg__min_data_per_group': 120}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.06646899627959202, 'reg__num_leaves': 58, 'reg__subsample': 0.6710099259029736, 'reg__colsample_bytree': 0.6181759688280724, 'reg__min_child_samples': 19, 'reg__reg_alpha': 3.77068862171459, 'reg__reg_lambda': 41.173184386793295, 'reg__min_split_gain': 3.9904241316470035, 'reg__path_smooth': 4.4069514657886195, 'reg__min_child_weight': 0.004085361457998897, 'reg__feature_fraction': 0.5442257912212237, 'reg__bagging_fraction': 0.8971450190436614, 'reg__bagging_freq': 3, 'reg__max_bin': 82, 'reg__min_data_per_group': 122}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.06422237141458872, 'reg__num_leaves': 58, 'reg__subsample': 0.7365158176517559, 'reg__colsample_bytree': 0.5959988140665754, 'reg__min_child_samples': 21, 'reg__reg_alpha': 3.8331434630237866, 'reg__reg_lambda': 39.81995825867953, 'reg__min_split_gain': 4.17299077831461, 'reg__path_smooth': 2.084443583466315, 'reg__min_child_weight': 0.00407422734506822, 'reg__feature_fraction': 0.5486201820055796, 'reg__bagging_fraction': 0.8946049604089042, 'reg__bagging_freq': 3, 'reg__max_bin': 83, 'reg__min_data_per_group': 113}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.06346940592426803, 'reg__num_leaves': 52, 'reg__subsample': 0.6754913618288426, 'reg__colsample_bytree': 0.6052921411906673, 'reg__min_child_samples': 19, 'reg__reg_alpha': 2.9424936743659655, 'reg__reg_lambda': 43.116636757485544, 'reg__min_split_gain': 3.920063003788643, 'reg__path_smooth': 2.0741518170055095, 'reg__min_child_weight':

0.004043711229263104, 'reg__feature_fraction': 0.5428026350591357, 'reg__bagging_fraction': 0.8248309365625599, 'reg__bagging_freq': 2, 'reg__max_bin': 83, 'reg__min_data_per_group': 118}

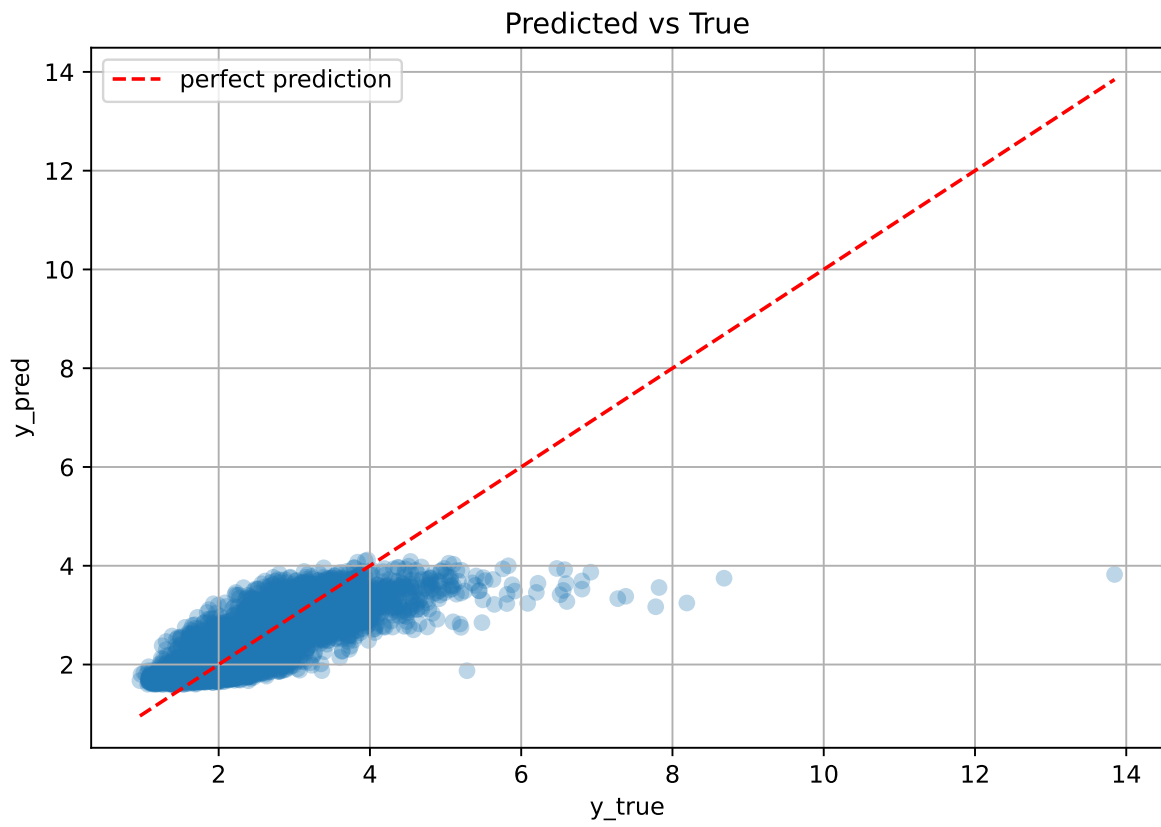
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.06056626644176634, 'reg__num_leaves': 44, 'reg__subsample': 0.6679958947913801, 'reg__colsample_bytree': 0.6761617551489637, 'reg__min_child_samples': 24, 'reg__reg_alpha': 3.7741541735562825, 'reg__reg_lambda': 30.909529834472863, 'reg__min_split_gain': 4.116990915898055, 'reg__path_smooth': 4.390949820574756, 'reg__min_child_weight': 0.0028347241131755027, 'reg__feature_fraction': 0.5490310345325161, 'reg__bagging_fraction': 0.8278958463222825, 'reg__bagging_freq': 3, 'reg__max_bin': 94, 'reg__min_data_per_group': 148}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.015043076668715441, 'reg__num_leaves': 58, 'reg__subsample': 0.7542399263911226, 'reg__colsample_bytree': 0.6748464615253584, 'reg__min_child_samples': 23, 'reg__reg_alpha': 3.6075189509092467, 'reg__reg_lambda': 43.54543010896449, 'reg__min_split_gain': 4.210943594484358, 'reg__path_smooth': 2.158562591627377, 'reg__min_child_weight': 0.0025997818122055418, 'reg__feature_fraction': 0.5387368475539817, 'reg__bagging_fraction': 0.8958781606213639, 'reg__bagging_freq': 3, 'reg__max_bin': 85, 'reg__min_data_per_group': 150}

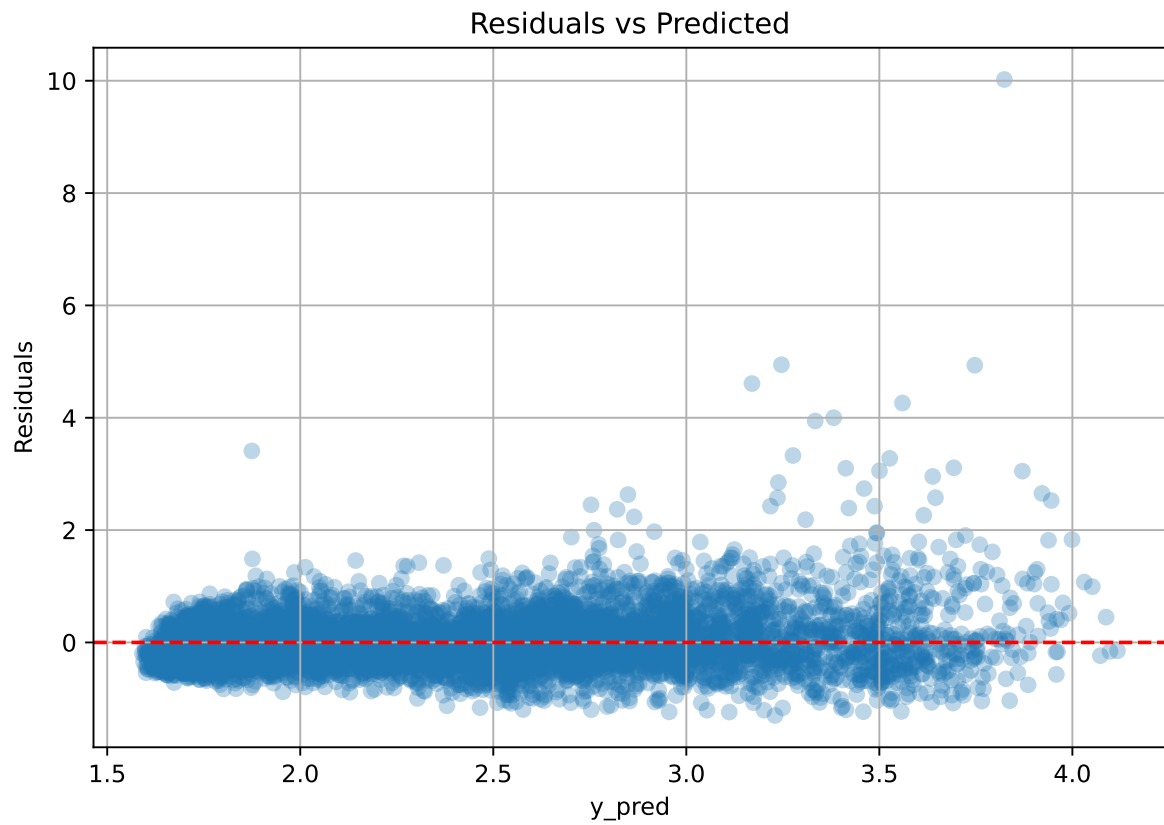
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.034758872585493455, 'reg__num_leaves': 55, 'reg__subsample': 0.6127166867231157, 'reg__colsample_bytree': 0.7939809622276951, 'reg__min_child_samples': 28, 'reg__reg_alpha': 3.258152119702933, 'reg__reg_lambda': 17.403314175408404, 'reg__min_split_gain': 4.437490763896928, 'reg__path_smooth': 2.3746522214744514, 'reg__min_child_weight': 0.0016480884693854878, 'reg__feature_fraction': 0.5325188969642721, 'reg__bagging_fraction': 0.7650113424074225, 'reg__bagging_freq': 3, 'reg__max_bin': 93, 'reg__min_data_per_group': 115}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.05922103338352456, 'reg__num_leaves': 58, 'reg__subsample': 0.7811469147710065, 'reg__colsample_bytree': 0.501610045011634, 'reg__min_child_samples': 25, 'reg__reg_alpha': 3.5710815597669354, 'reg__reg_lambda': 41.30555639518067, 'reg__min_split_gain': 4.239904150628457, 'reg__path_smooth': 2.2037999213337702, 'reg__min_child_weight': 0.002738332059976123, 'reg__feature_fraction': 0.5341717646656486, 'reg__bagging_fraction': 0.8995792623468389, 'reg__bagging_freq': 3, 'reg__max_bin': 85, 'reg__min_data_per_group': 118}

3.5.2 Scatter



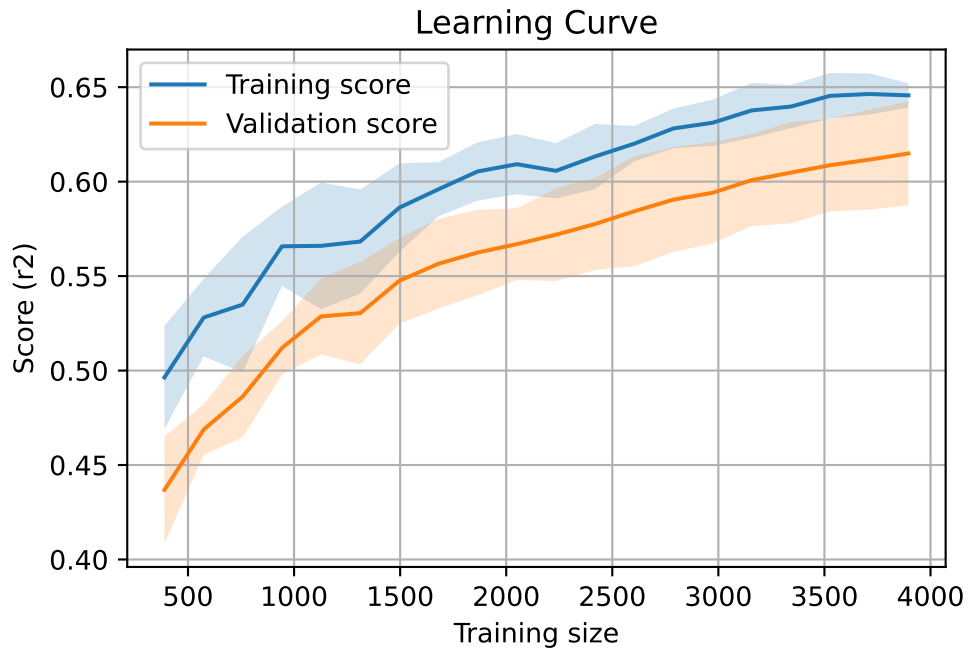
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7952, `Len(training)` : 5567,

`Len(training_real)` : 3897, `Len(test_of_training)` : 1670, `Len(test)` : 2385



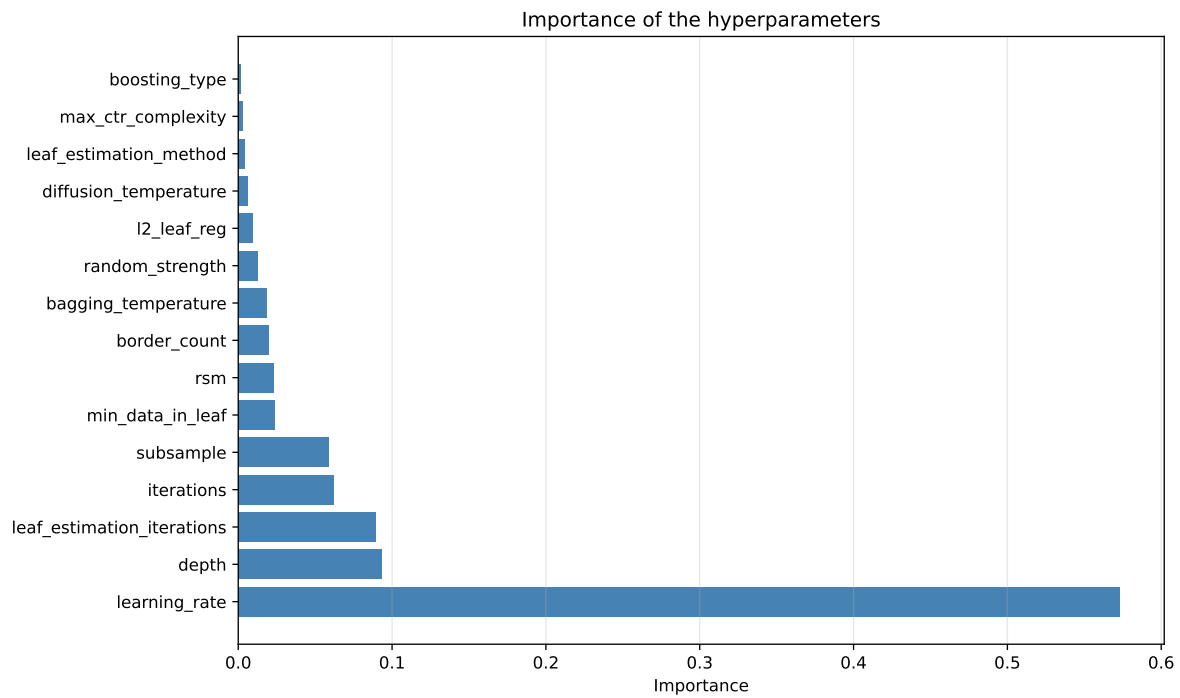
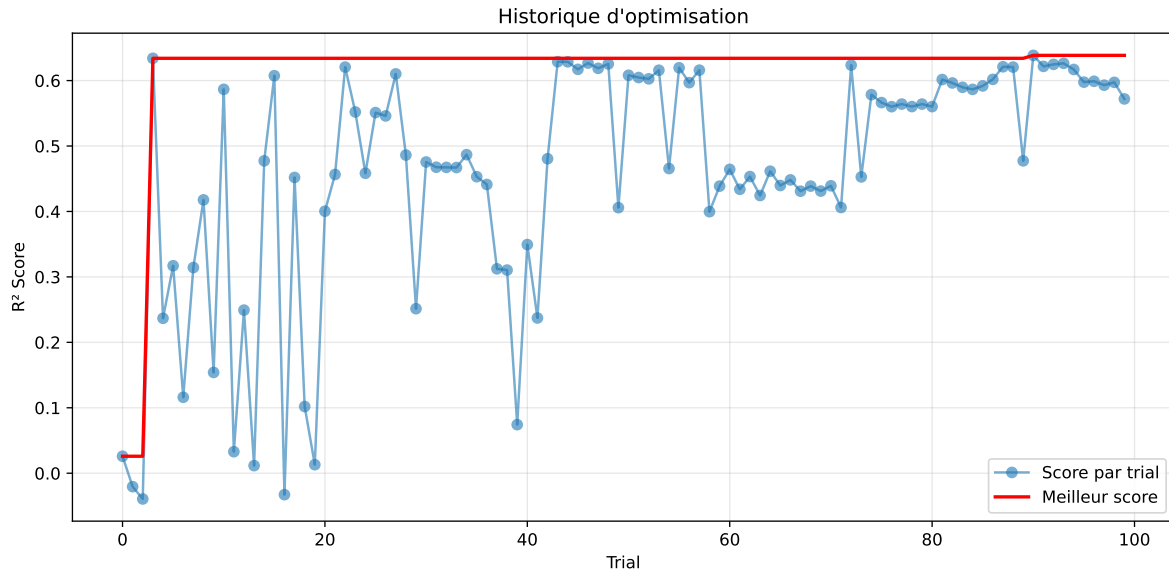
3.6 Model : CatBoost

3.6.1 Gridsearch

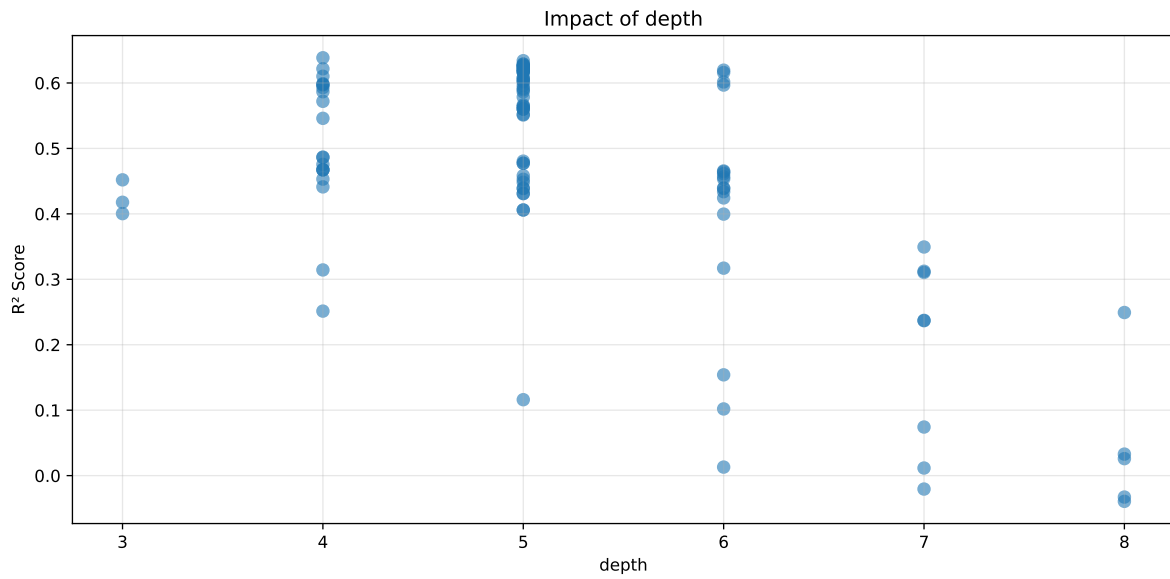
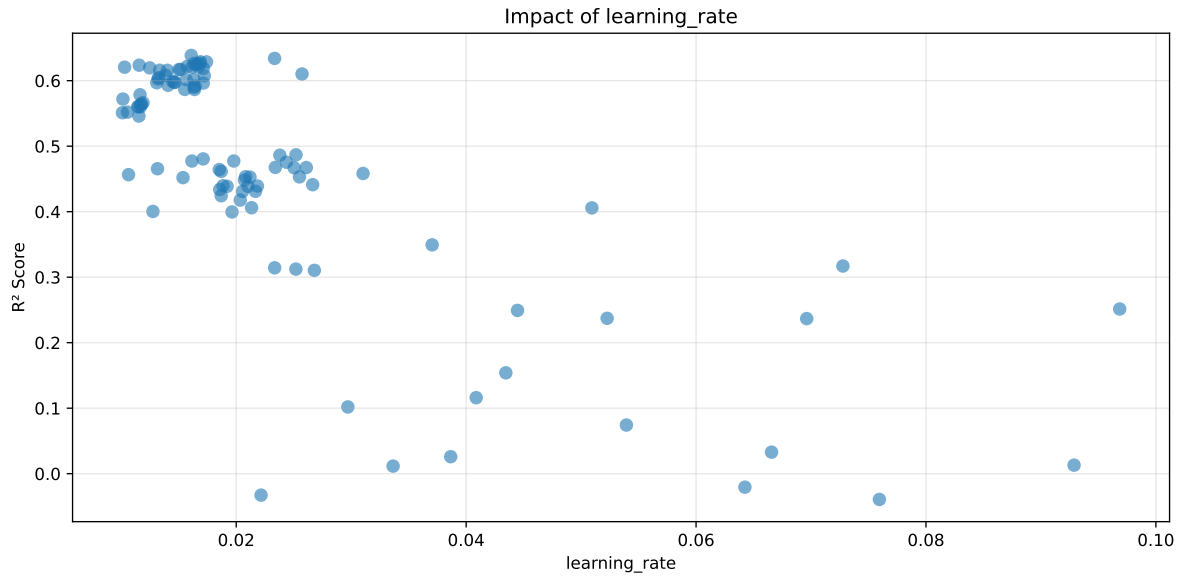
Distribution y_train vs y_test: y_train: mean=2.349, std=0.796, min=0.959, max=13.845
y_test: mean=2.366, std=0.808, min=1.011, max=7.822

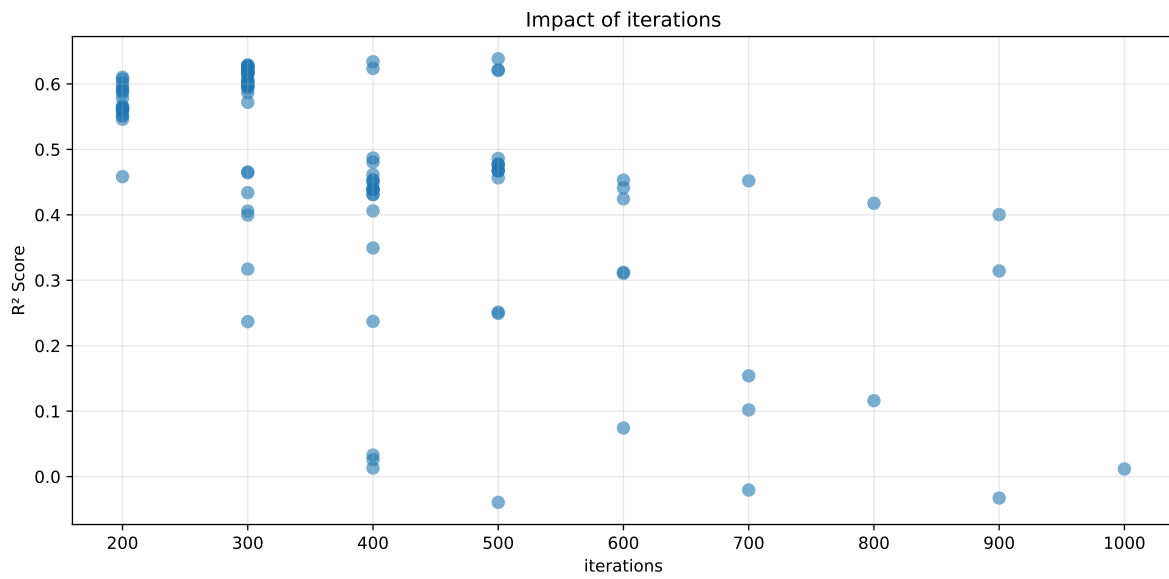
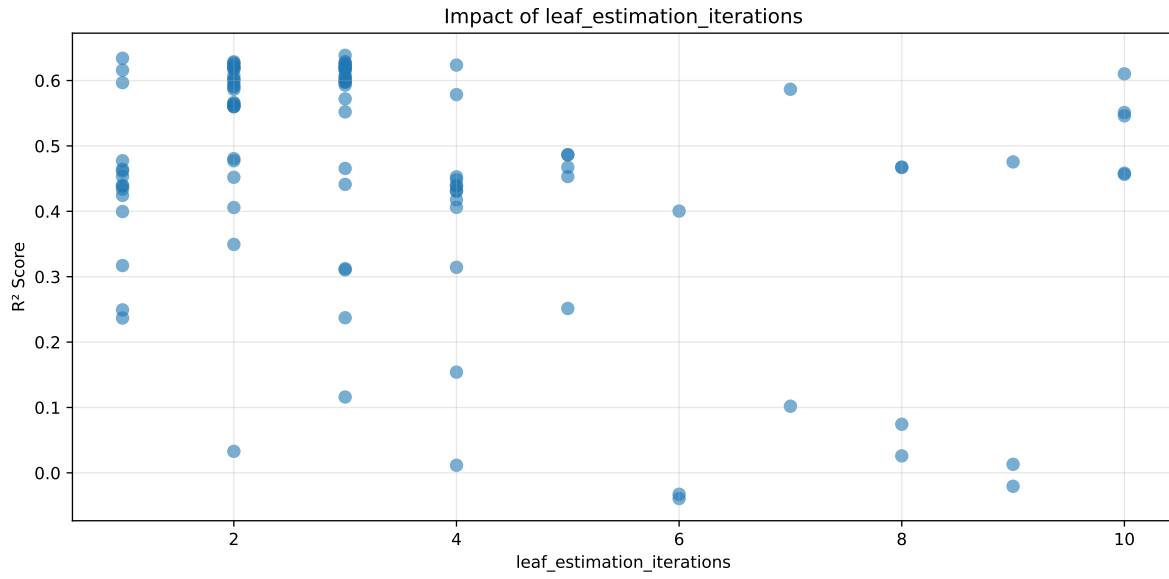
===== TOP Importance FEATURES =====

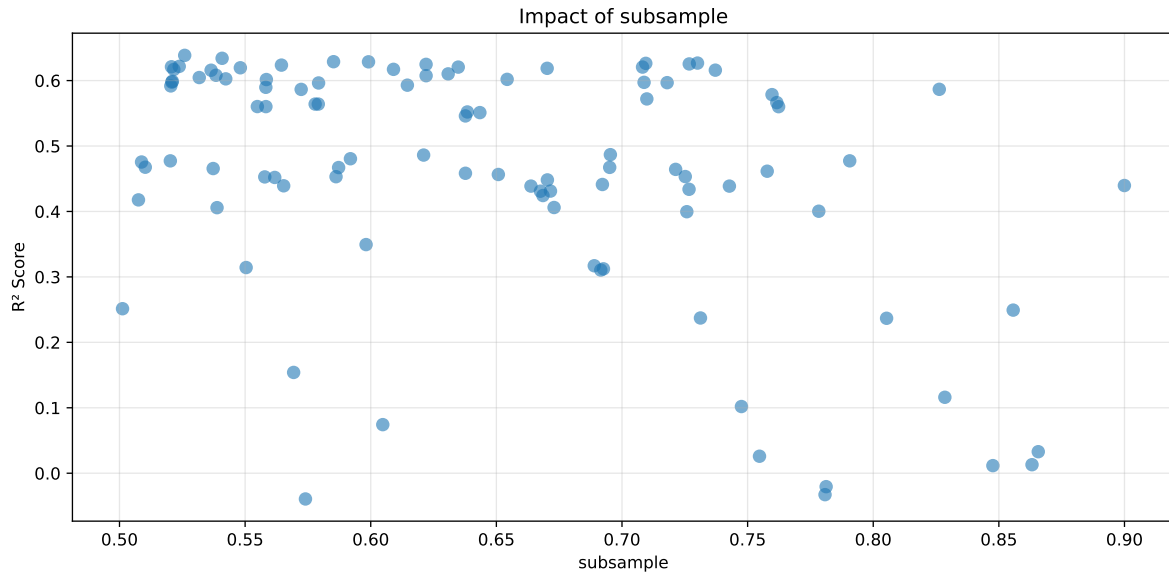
feature	importance	r_temperature_st_6h_diff_last	r_umidity
r_temperature_st_6h_diff_last	21.510017		19.515777
r_CO2	9.681841	r_temperature	7.733908
r_temperature_st_6h_max	6.049018	r_NH3_st_6h_max	5.181447
r_NH3_st_6h_max	5.175490	r_NH3	5.175490
r_NH3	5.175490	r_umidity_st_6h_diff_last	4.925604
r_umidity_st_6h_diff_last	4.925604	r_NH3_st_6h_std	4.800886
r_NH3_st_6h_std	4.800886	r_THI	3.444296
r_THI	3.444296	r_NH3_st_6h_slope	2.604865
r_NH3_st_6h_slope	2.604865	r_NH3_st_6h_diff_last	1.252151
r_NH3_st_6h_diff_last	1.252151	r_umidity_st_6h_std	0.792887
r_umidity_st_6h_std	0.792887		



Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'leaf_estimation_iterations', 'iterations', 'subsample']

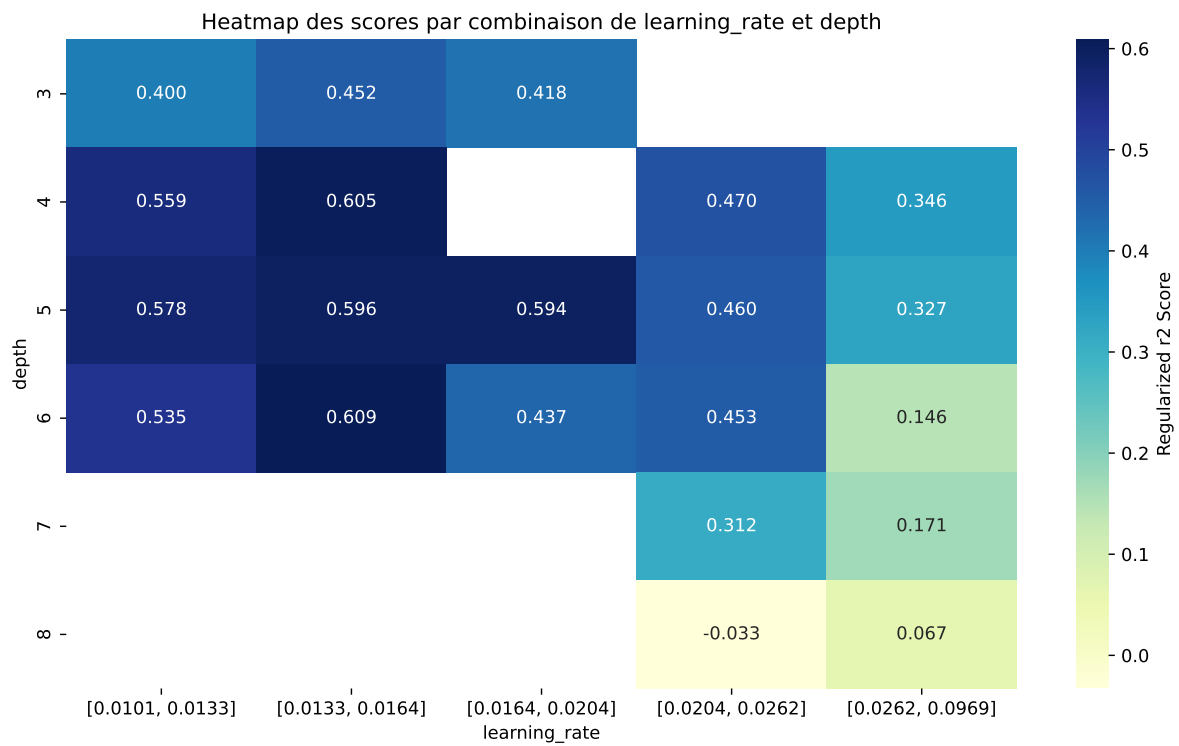






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6688	0.6385	0.636	0.0232	0.6385	0.6459	0.0074	1.0116	1.0474
2	0.6643	0.634	0.6359	0.0233	0.634	0.647	0.0129	1.0204	1.0477
3	0.6584	0.6288	0.627	0.0232	0.6288	0.6369	0.008	1.0127	1.047
4	0.6589	0.6287	0.6275	0.0255	0.6287	0.6377	0.009	1.0143	1.048
5	0.6557	0.6267	0.6241	0.0226	0.6267	0.6354	0.0087	1.0139	1.0463
6	0.6557	0.6263	0.6236	0.0232	0.6263	0.6339	0.0076	1.0122	1.047
7	0.6537	0.6254	0.6226	0.0222	0.6254	0.6357	0.0102	1.0164	1.0452
8	0.6546	0.6248	0.6211	0.023	0.6248	0.6326	0.0078	1.0126	1.0477
9	0.6534	0.6236	0.6224	0.0234	0.6236	0.6324	0.0087	1.014	1.0477
10	0.6486	0.6217	0.6179	0.0222	0.6217	0.6271	0.0054	1.0087	1.0433
---	---	---	---	---	---	---	---	---	---
100	0.9193	0.7595	0.7624	0.0355	-0.0393	0.7825	0.0231	1.0304	1.2104

Hyperparameters:

Rank 1: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.016087075249486275, 'reg__l2_leaf_reg': 47.04095406478758, 'reg__bagging_temperature': 1.1207813774235393, 'reg__random_strength': 1.8484646725193423, 'reg__subsample': 0.5259490115934964, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5065723479560909, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4842.7541810278435}

Rank 2: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.02333753024620632, 'reg__l2_leaf_reg': 35.64115945135981, 'reg__bagging_temperature': 4.347379715736608, 'reg__random_strength': 3.9046099401423113, 'reg__subsample': 0.5408443380301956, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.5022494084431961, 'reg__border_count': 107, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7441.014471082471}

Rank 3: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.016886248390815577, 'reg__l2_leaf_reg': 38.6229531915675, 'reg__bagging_temperature': 1.3497445072819731, 'reg__random_strength': 1.3360329977388368, 'reg__subsample': 0.5851980746347983, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5448399038327782, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 7080.160847115136}

Rank 4: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.017416926717651483, 'reg__l2_leaf_reg': 38.41620396959944, 'reg__bagging_temperature': 1.4118112224332875, 'reg__random_strength': 1.229148368582774, 'reg__subsample': 0.5991134986390579, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':

0.6154441852535613, 'reg__border_count': 86, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8964.123040272443}

Rank 5: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.016853810805283073, 'reg__l2_leaf_reg': 29.471928365714938, 'reg__bagging_temperature': 1.481295186249498, 'reg__random_strength': 1.3942110369483272, 'reg__subsample': 0.730056262404308, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.5276637495234594, 'reg__border_count': 89, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5293.994512722039}

Rank 6: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.016330859598144624, 'reg__l2_leaf_reg': 28.347921916749513, 'reg__bagging_temperature': 1.1594413443029958, 'reg__random_strength': 1.894183416463671, 'reg__subsample': 0.7094712414055101, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.578756910751075, 'reg__border_count': 54, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4808.296743428161}

Rank 7: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.016635180806181135, 'reg__l2_leaf_reg': 29.262612706327612, 'reg__bagging_temperature': 1.3881645932089273, 'reg__random_strength': 1.4699638173708307, 'reg__subsample': 0.7267724682144905, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6097039785113456, 'reg__border_count': 49, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5168.718836770716}

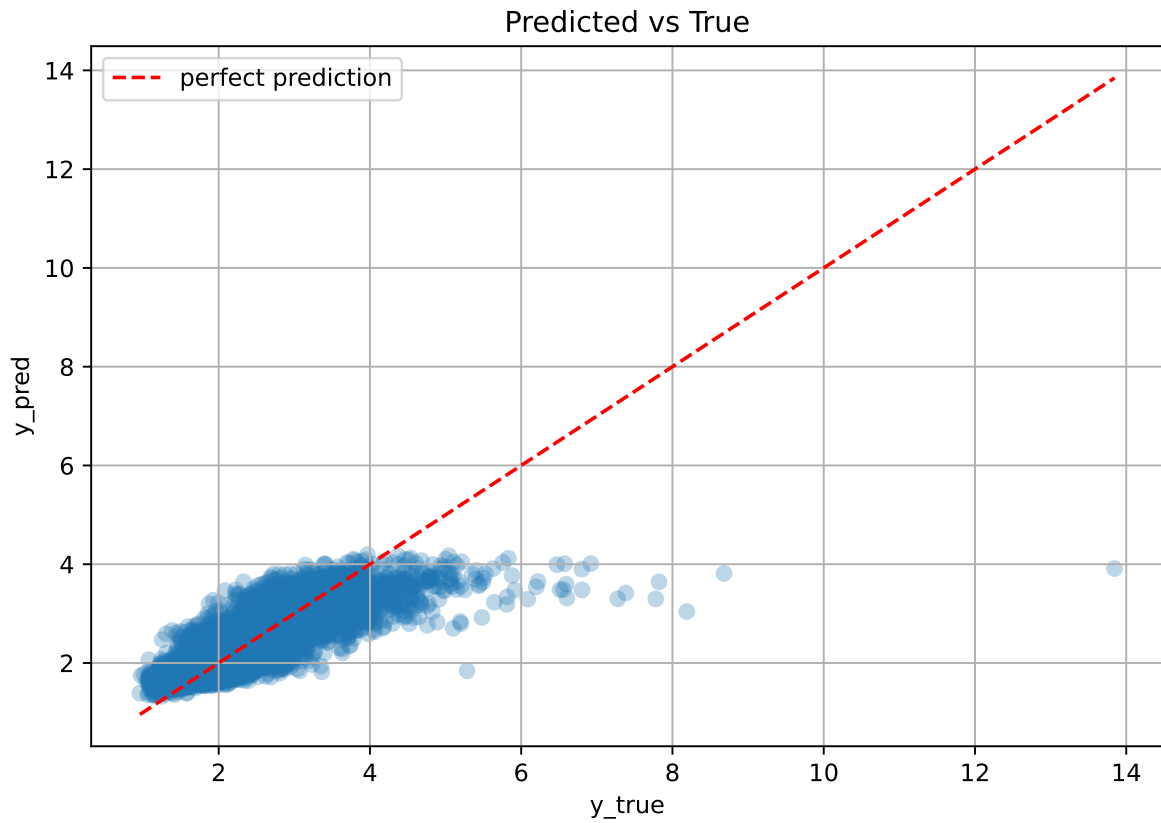
Rank 8: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.016512628273399672, 'reg__l2_leaf_reg': 27.598471363858348, 'reg__bagging_temperature': 1.1139778986960849, 'reg__random_strength': 1.8201988020665334, 'reg__subsample': 0.6220243115463021, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 3, 'reg__rsm': 0.5037219654496877, 'reg__border_count': 55, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 4817.320614615904}

Rank 9: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.011561228866561708, 'reg__l2_leaf_reg': 30.149515631435452, 'reg__bagging_temperature': 2.034340153318365, 'reg__random_strength': 1.6061258443241866, 'reg__subsample': 0.5644575518971933, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.5226573836398632, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5578.363498685761}

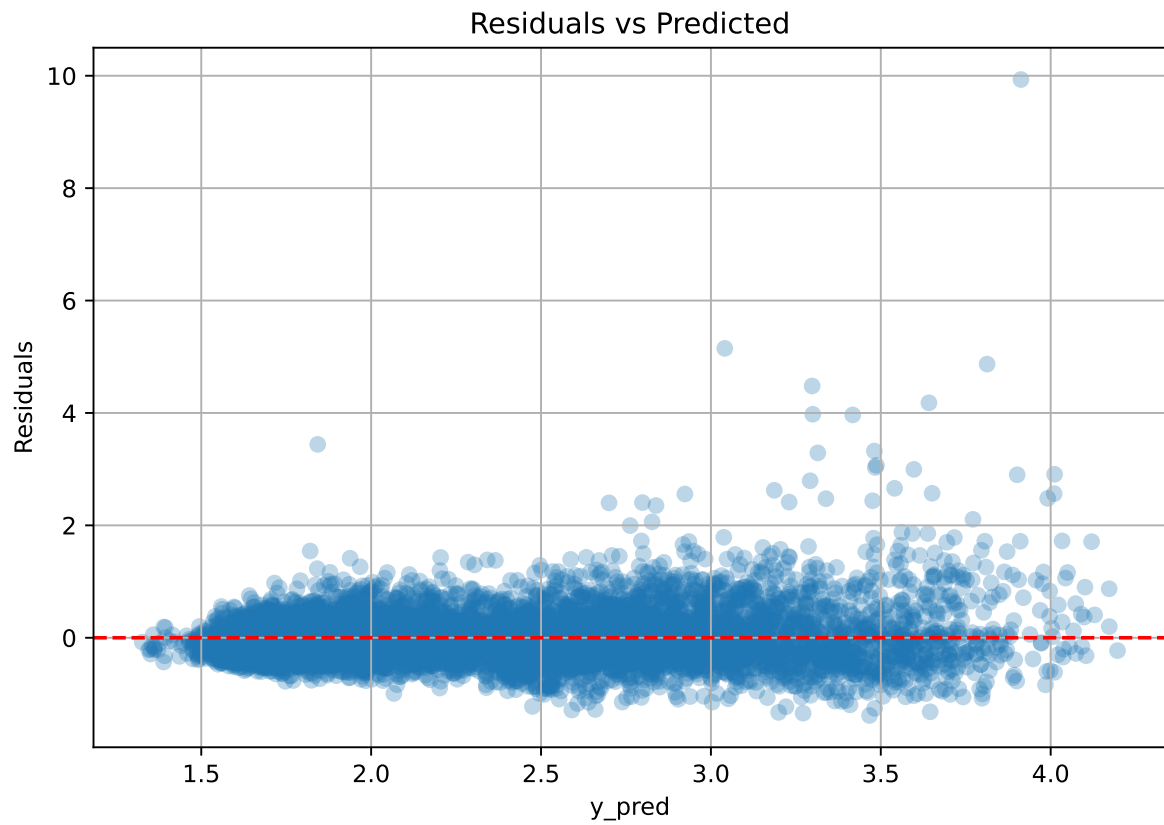
Rank 10: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.015736282169150306, 'reg__l2_leaf_reg': 27.6866941221477, 'reg__bagging_temperature': 1.1195311144559708,

```
'reg__random_strength': 4.174687150203805, 'reg__subsample': 0.5237742421498058,  
'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 3, 'reg__rsm':  
0.5030132844358228, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Gradient',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_tempera-  
ture': 4801.302283165839}
```

3.6.2 Scatter



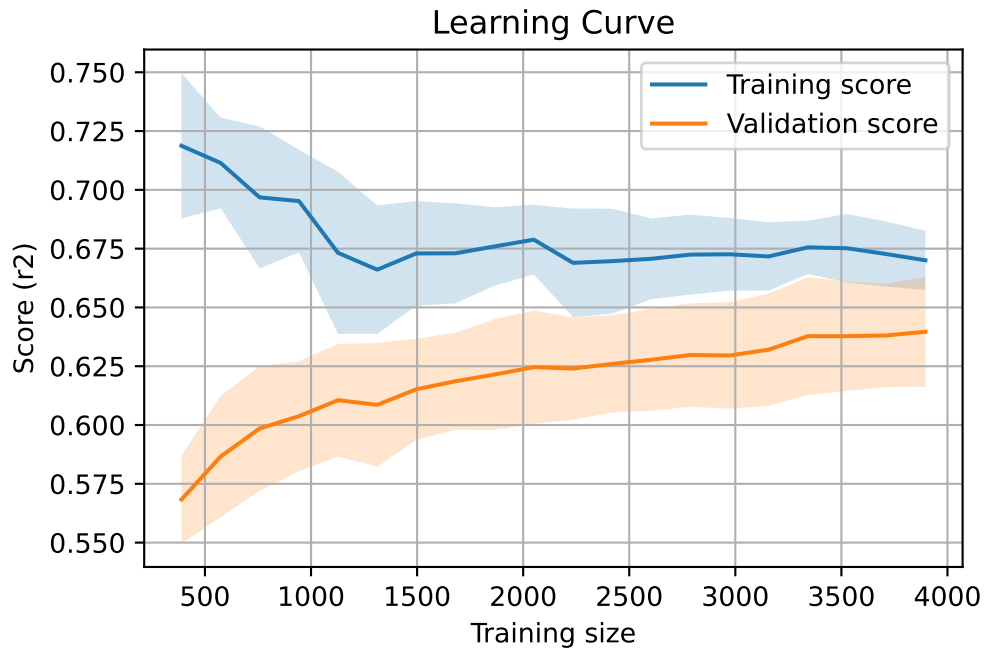
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7952, `Len(training)` : 5567,

`Len(training_real)` : 3897, `Len(test_of_training)` : 1670, `Len(test)` : 2385



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.507	0.49	0.014	0.482	0.483	-0.007	0.985	1.034	nan
LGBM	0.644	0.614	0.022	0.606	0.628	0.014	1.022	1.05	0.6138
CatBoost	0.669	0.638	0.023	0.636	0.646	0.007	1.012	1.047	0.6385

4 Plot

4.1 Standalone plots

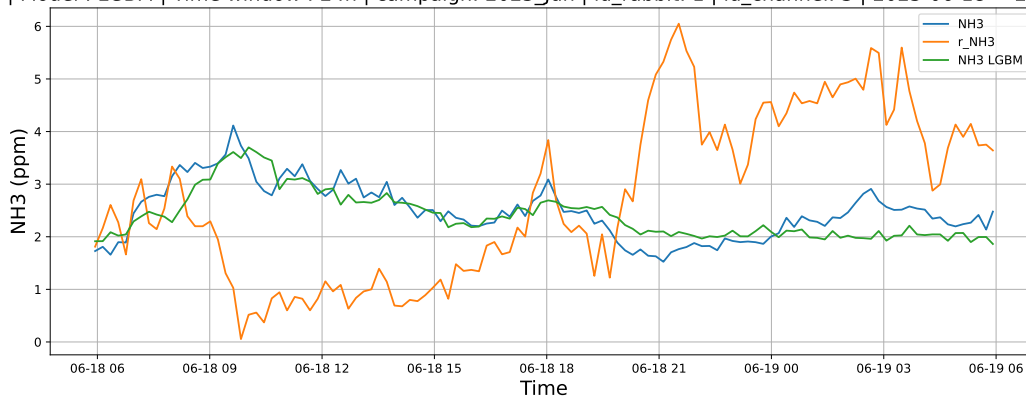
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

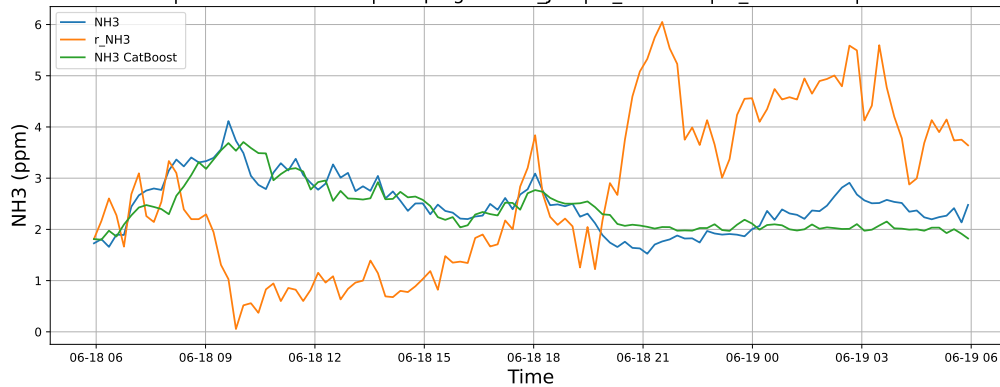
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



4.1.1.2 Time window : 48

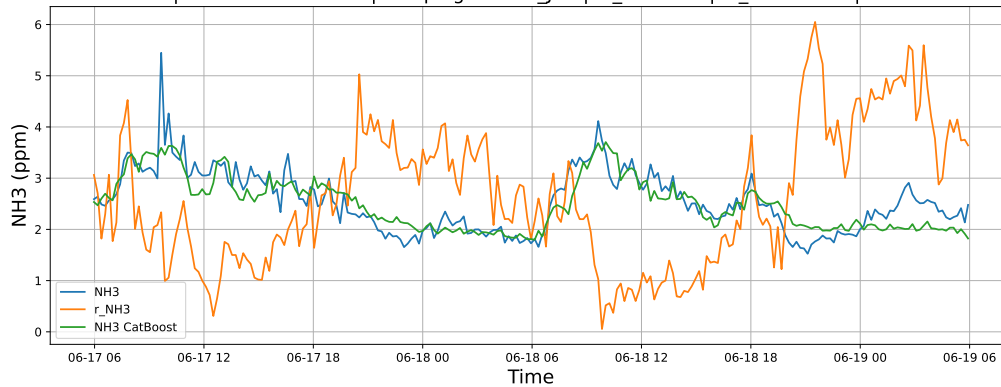
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

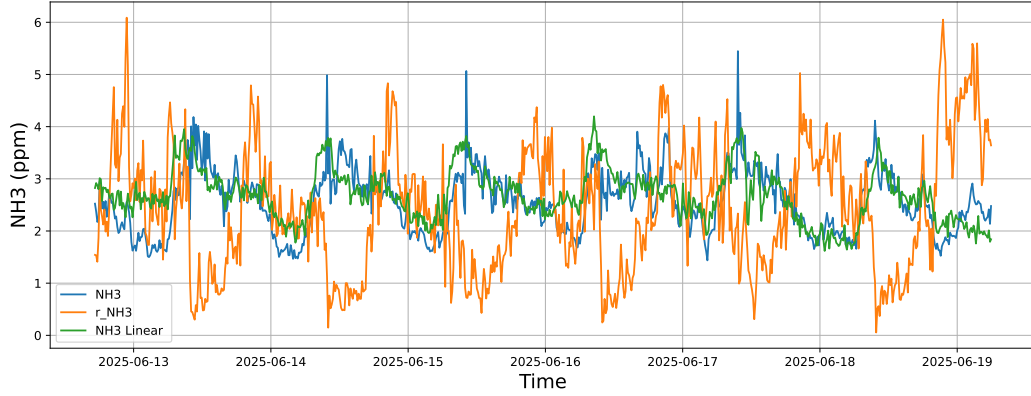


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

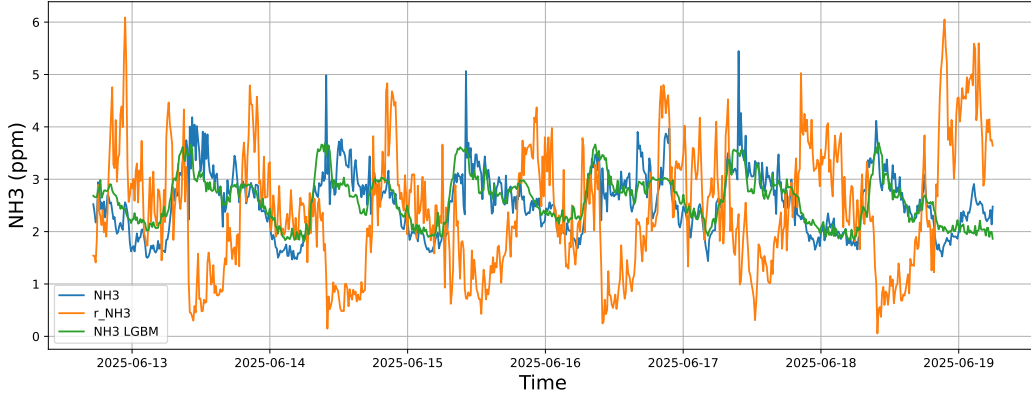


4.1.1.3 Time window : ALL

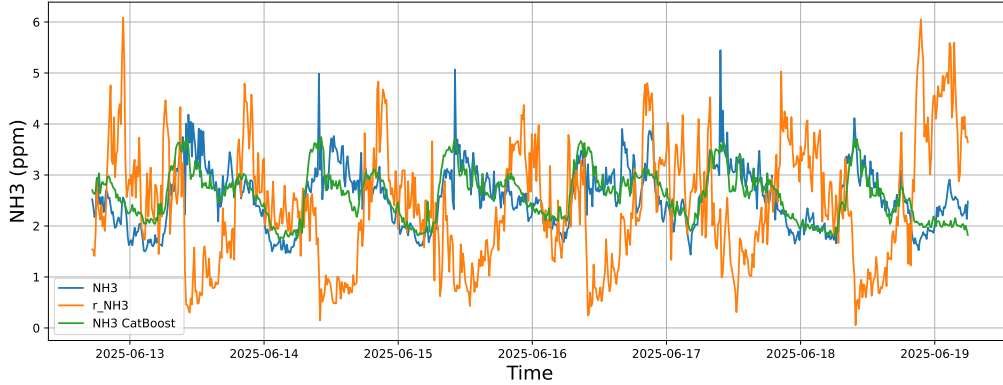
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



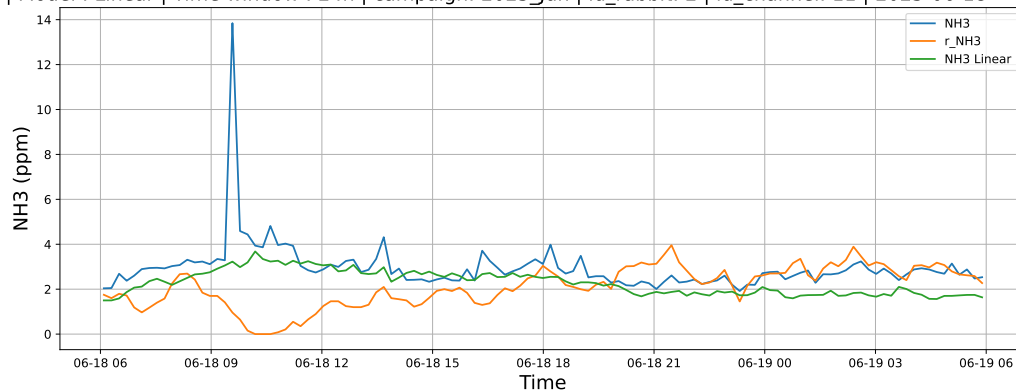
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



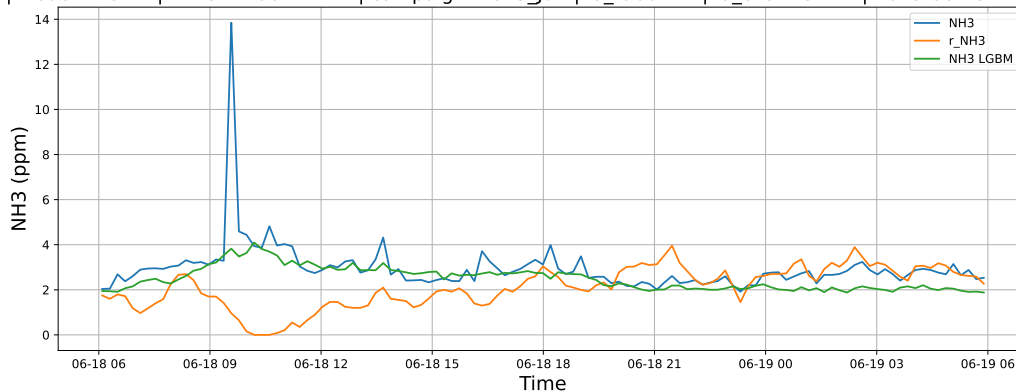
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

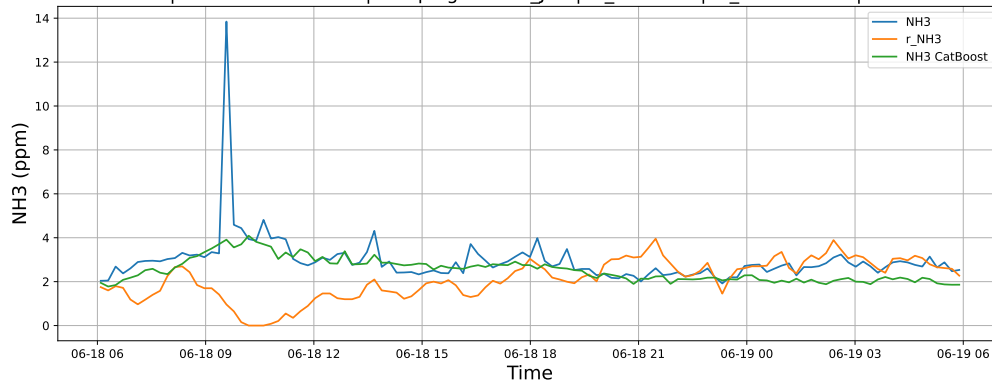
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

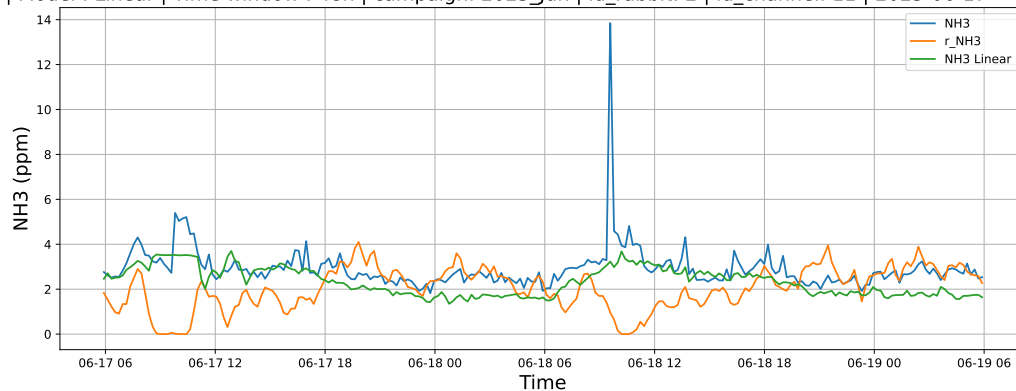


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

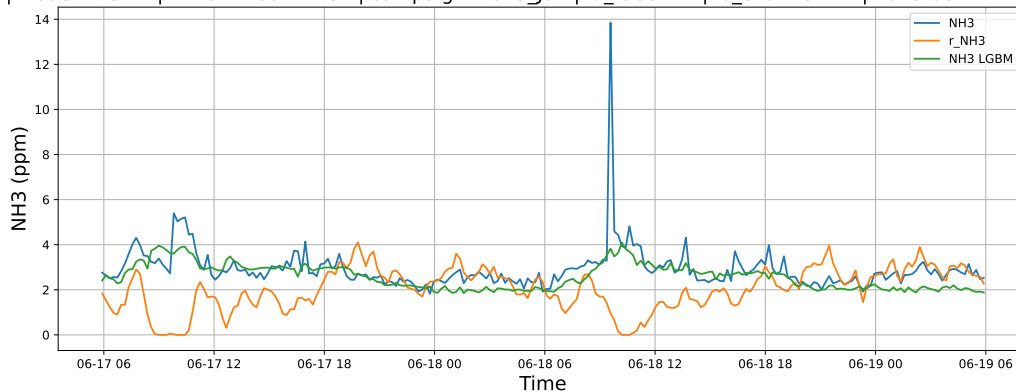


4.1.2.2 Time window : 48

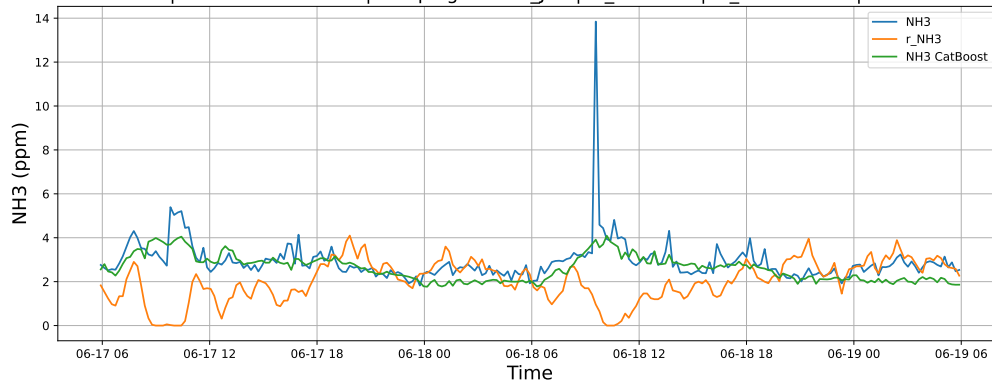
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

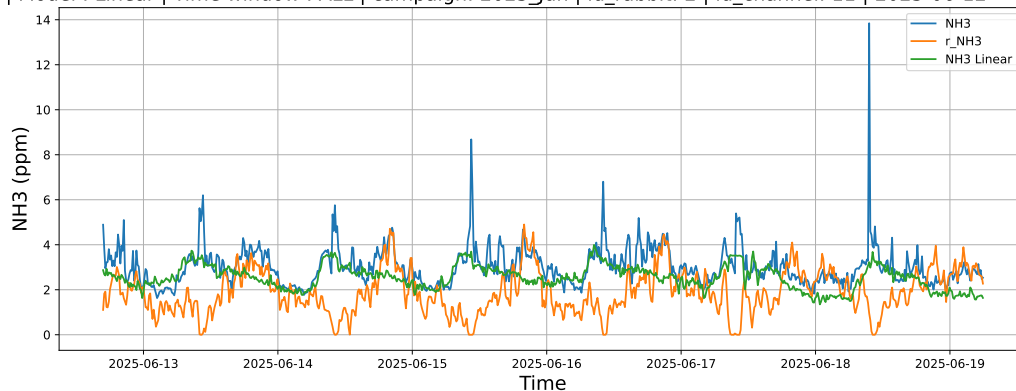


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

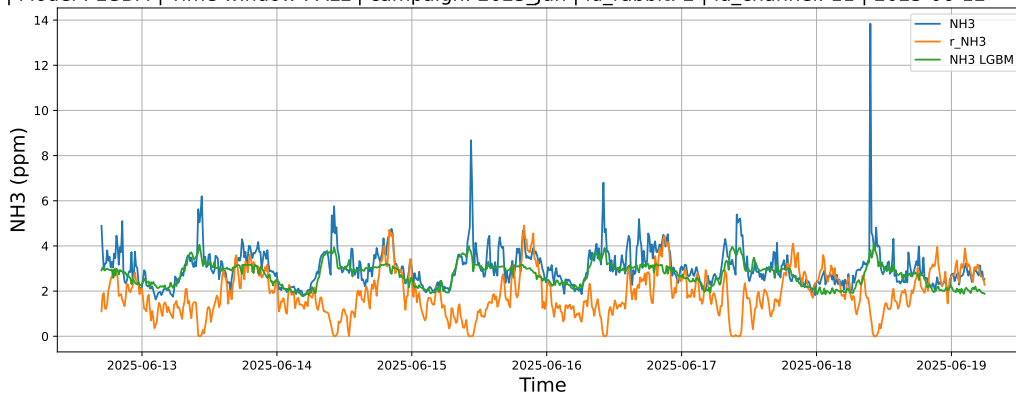


4.1.2.3 Time window : ALL

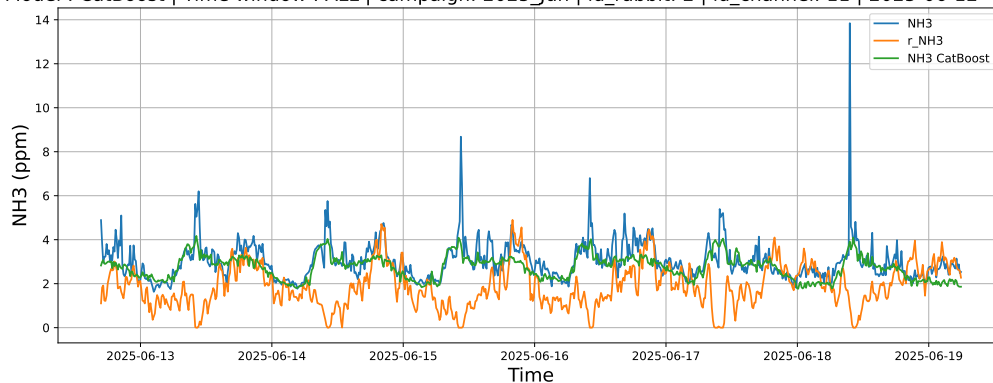
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



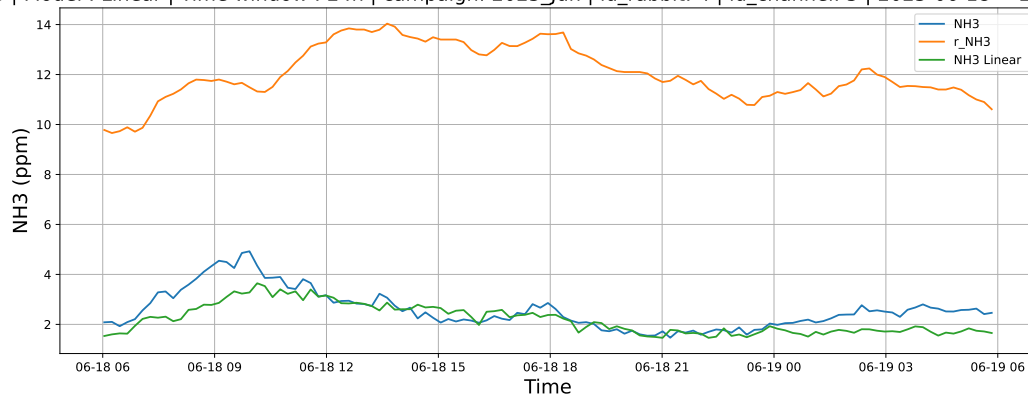
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



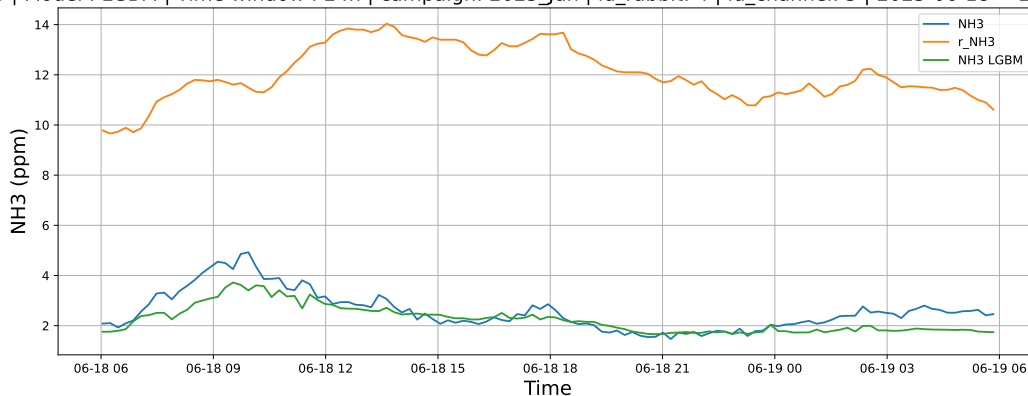
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

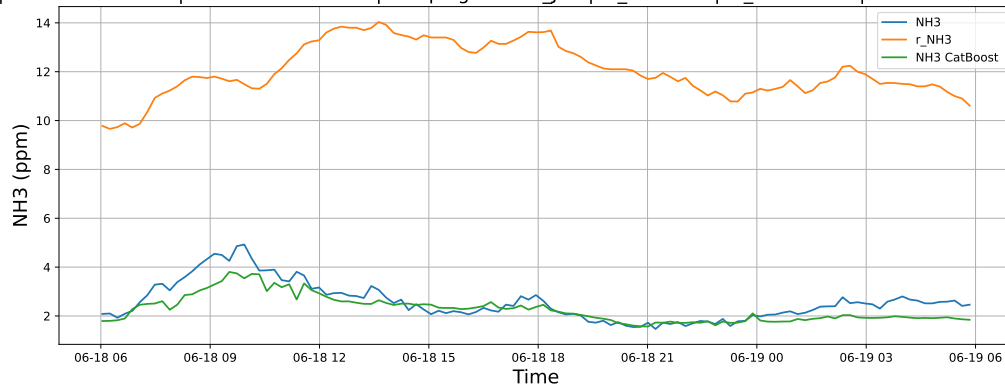
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

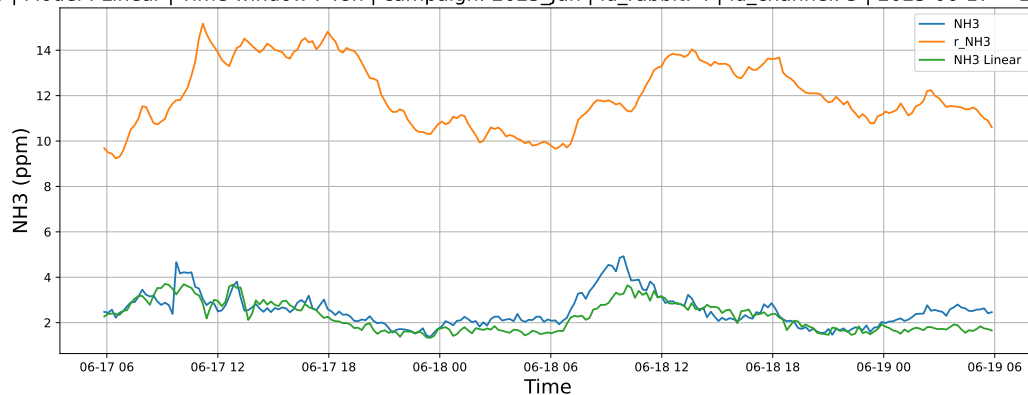


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

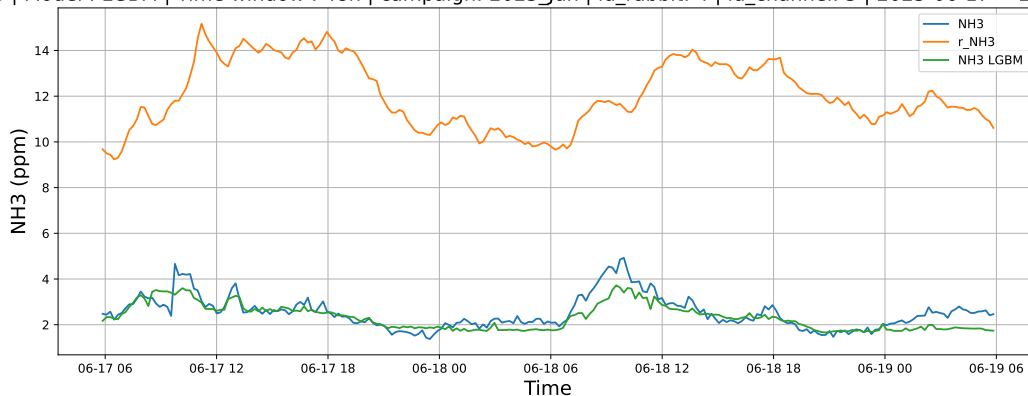


4.1.3.2 Time window : 48

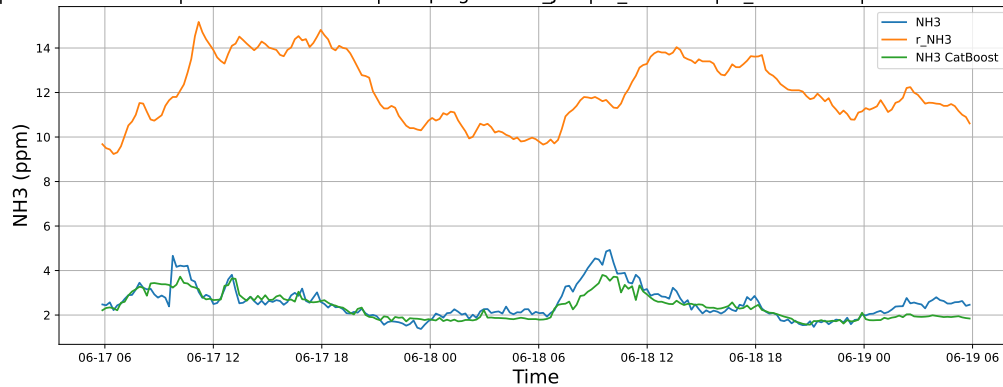
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

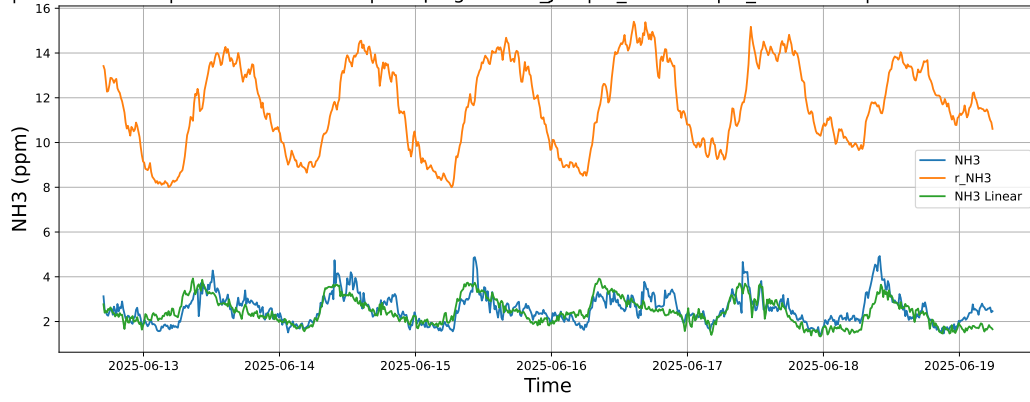


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

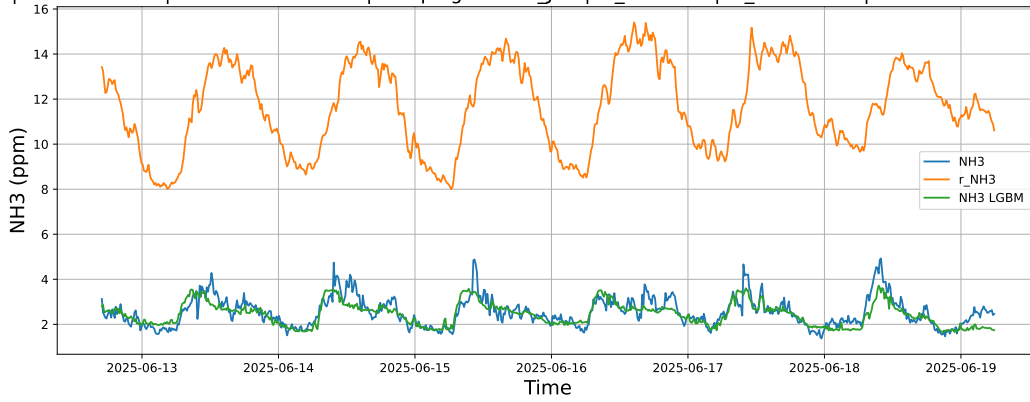


4.1.3.3 Time window : ALL

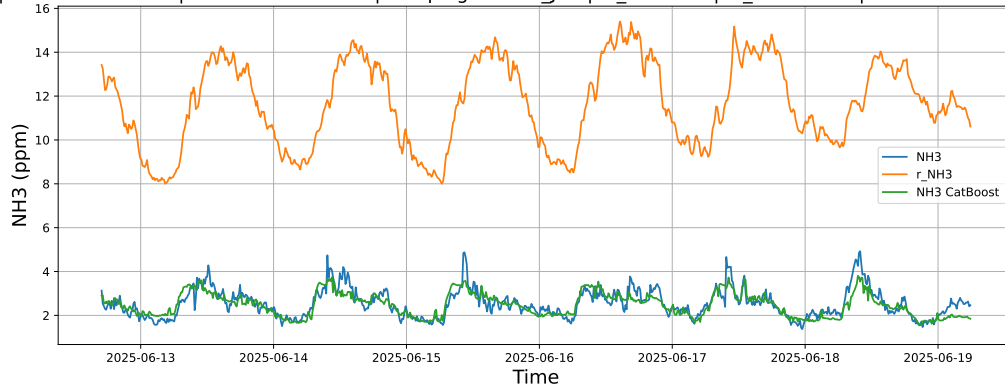
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



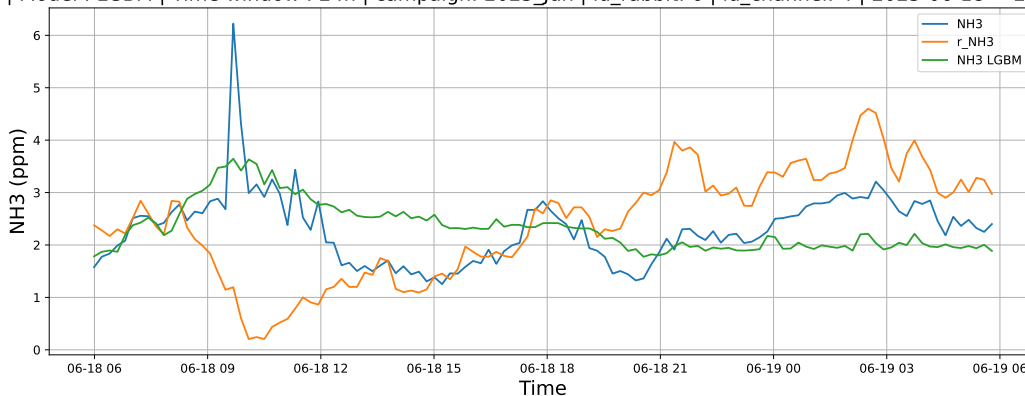
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

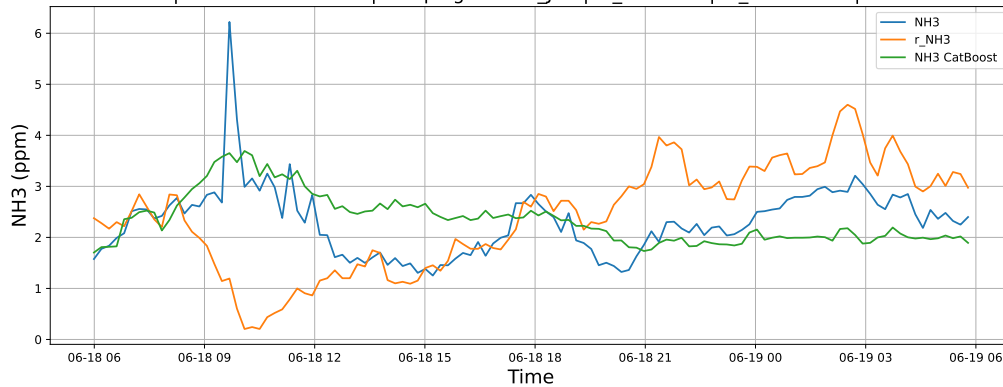
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

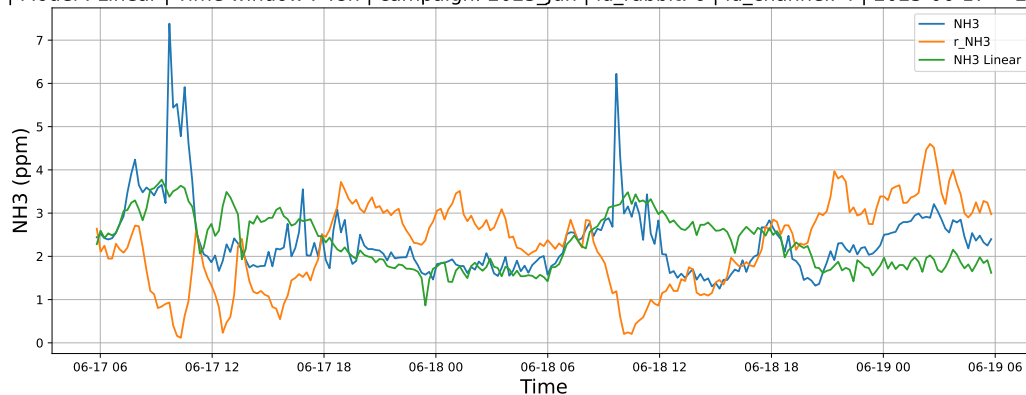


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

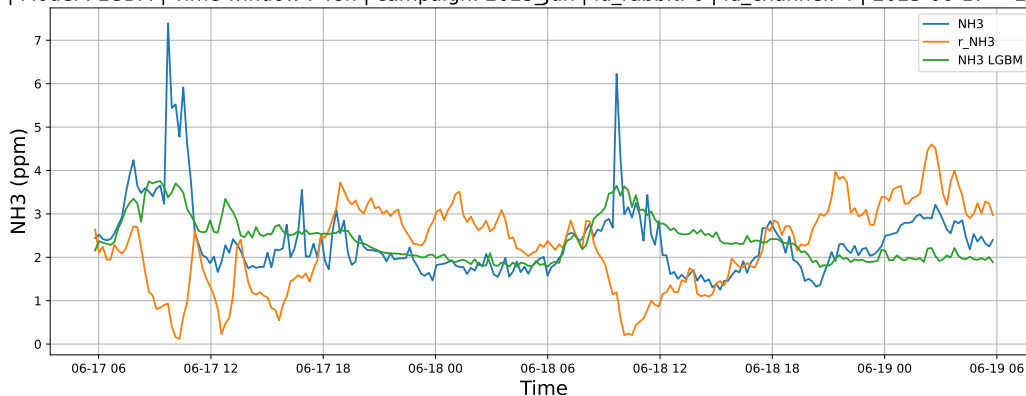


4.1.4.2 Time window : 48

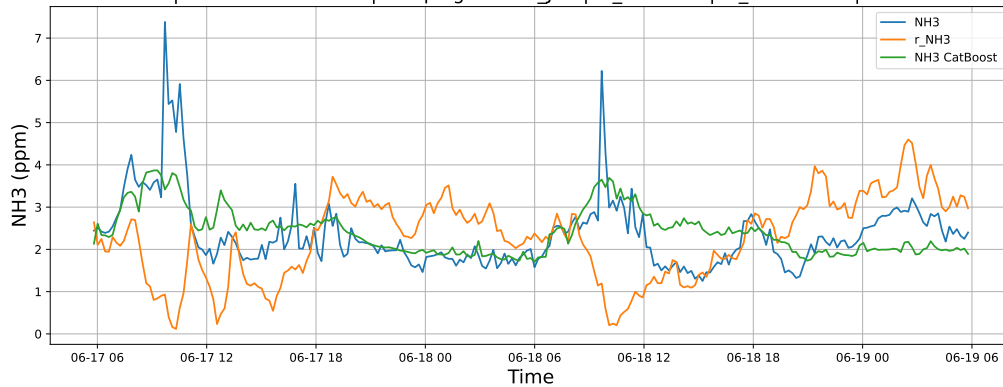
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

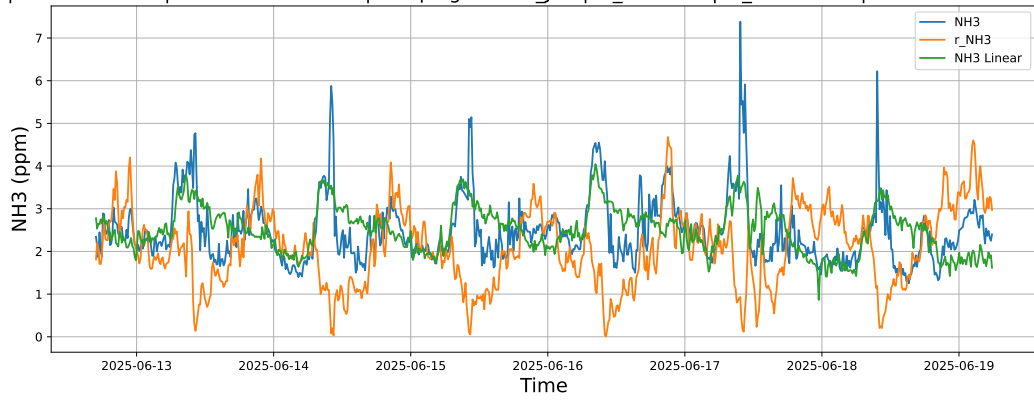


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

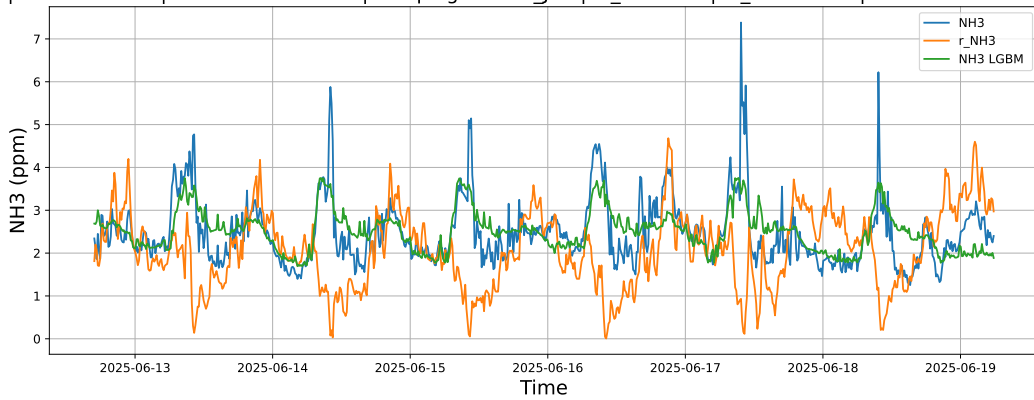


4.1.4.3 Time window : ALL

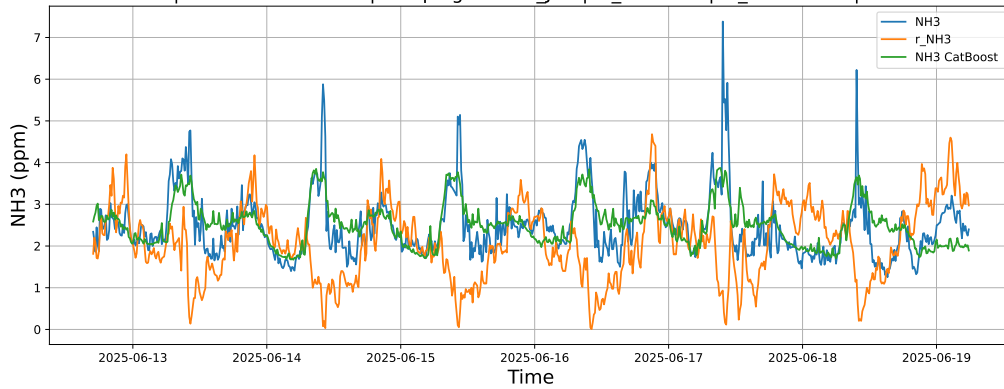
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



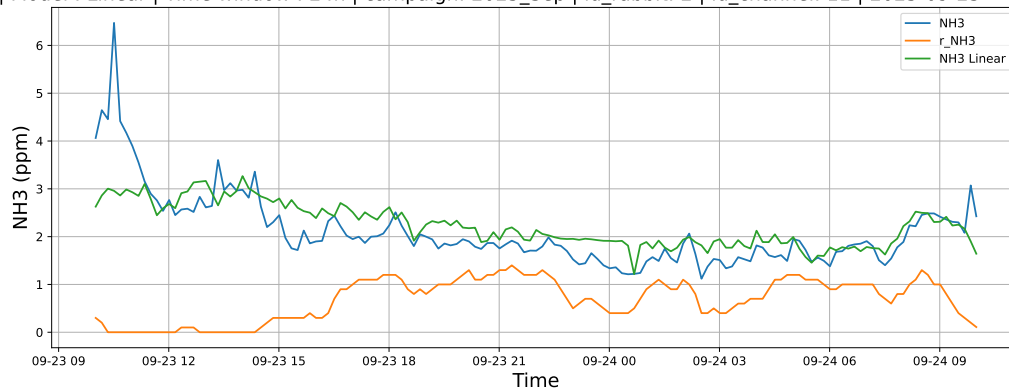
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



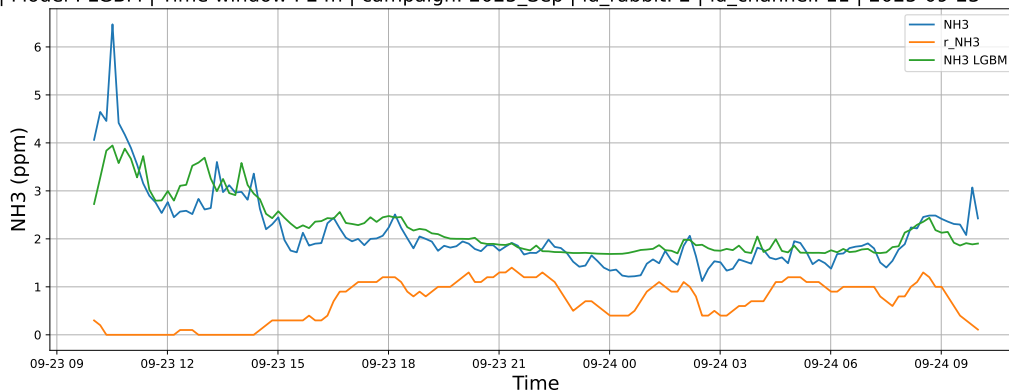
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

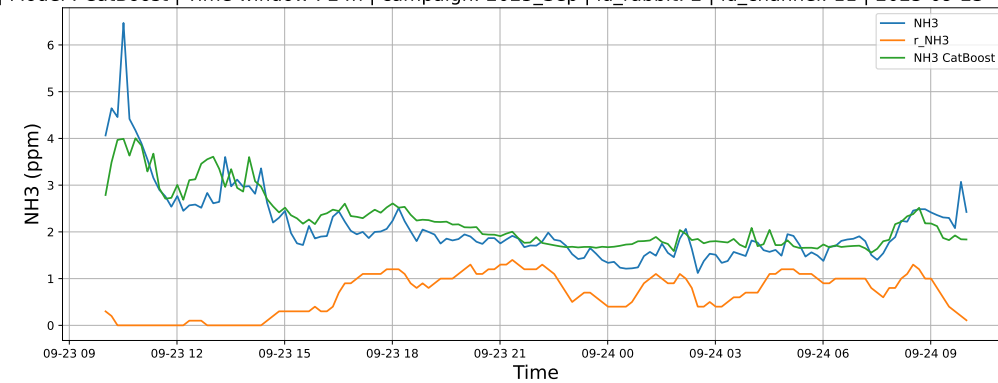
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

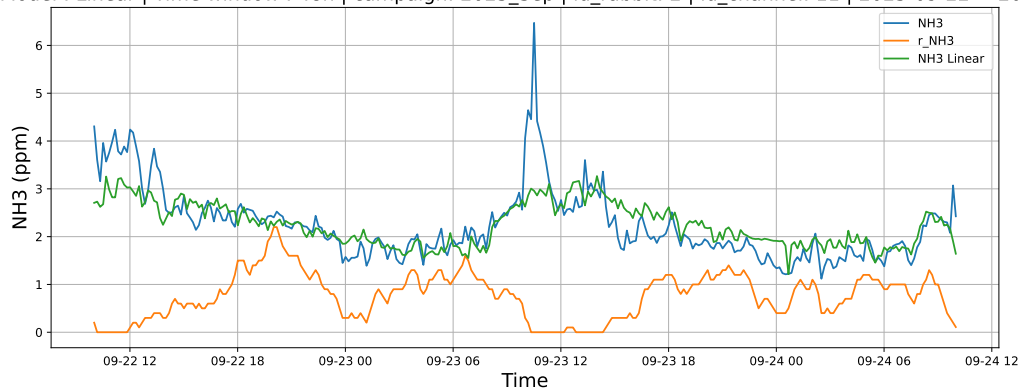


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

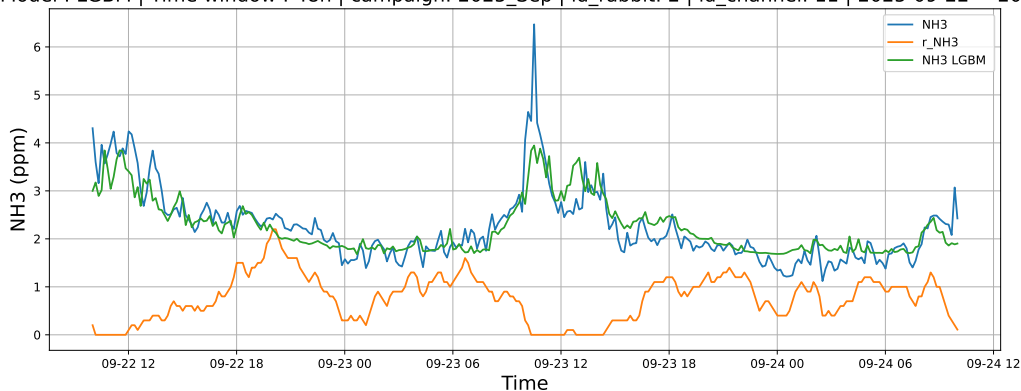


4.1.5.2 Time window : 48

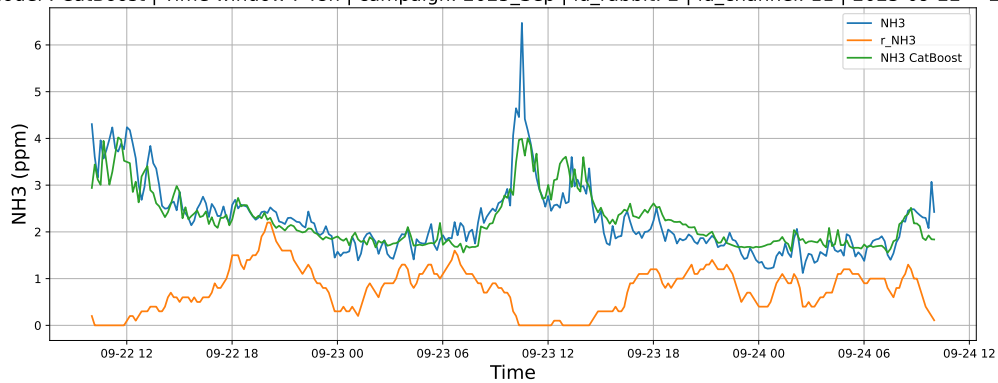
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

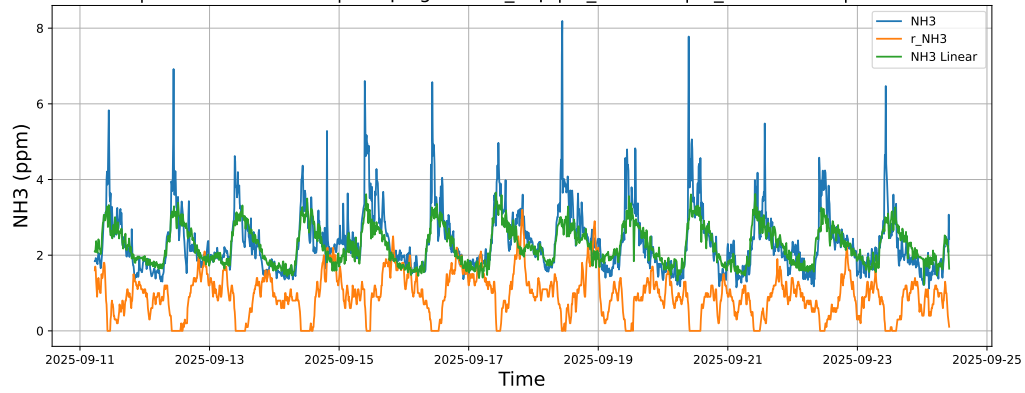


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

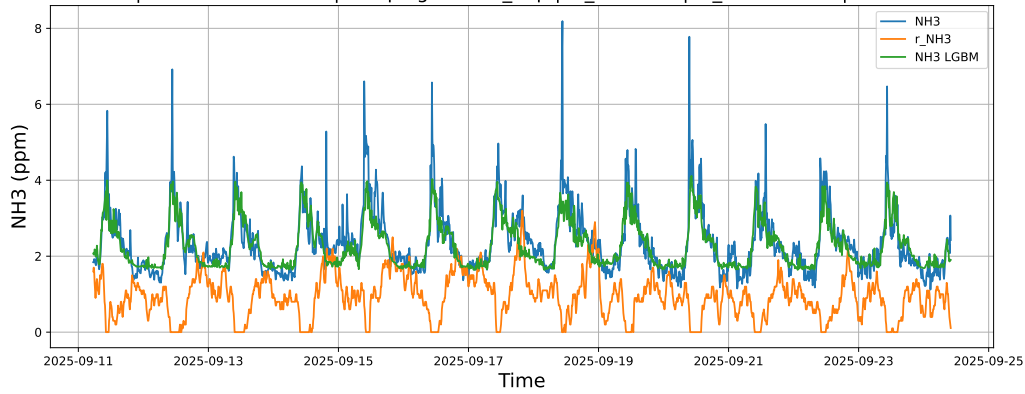


4.1.5.3 Time window : ALL

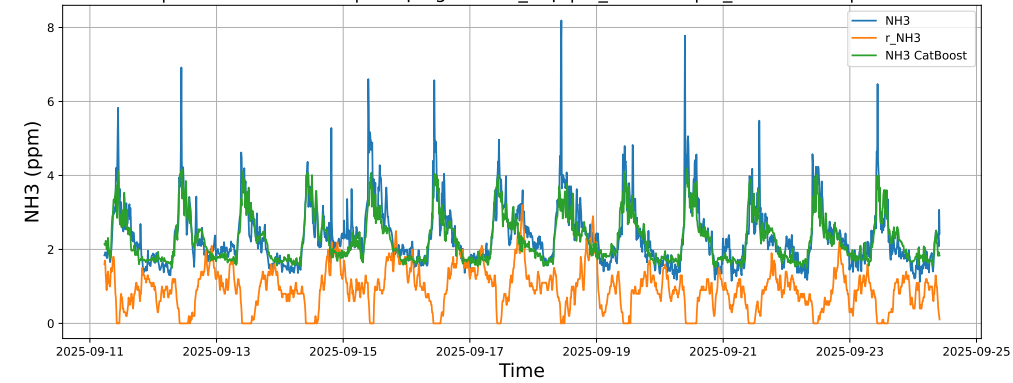
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



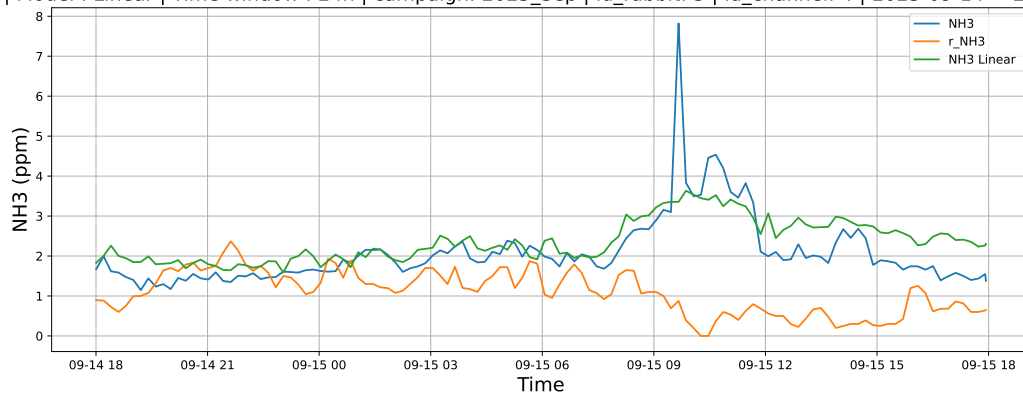
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



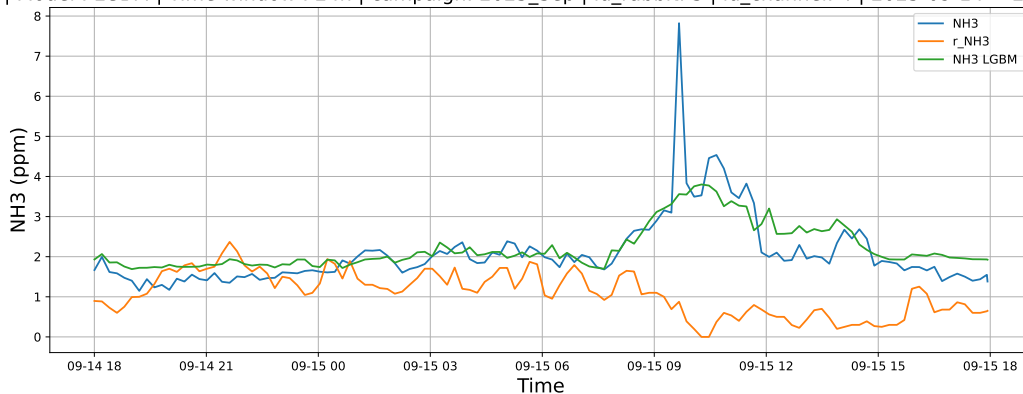
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

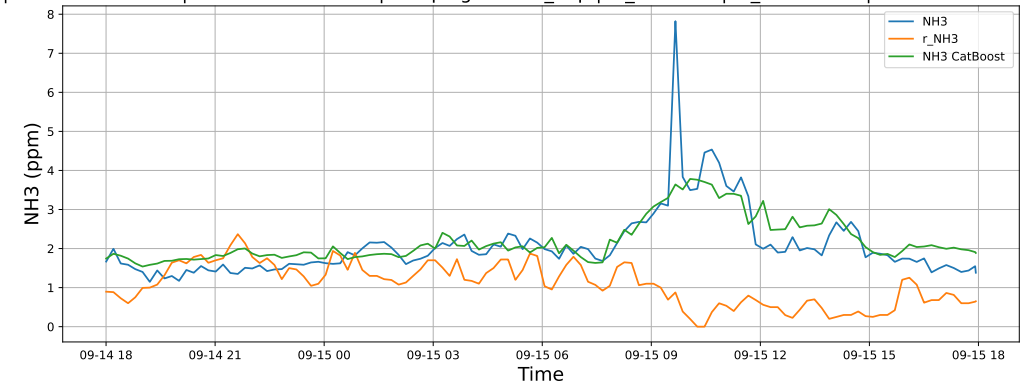
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

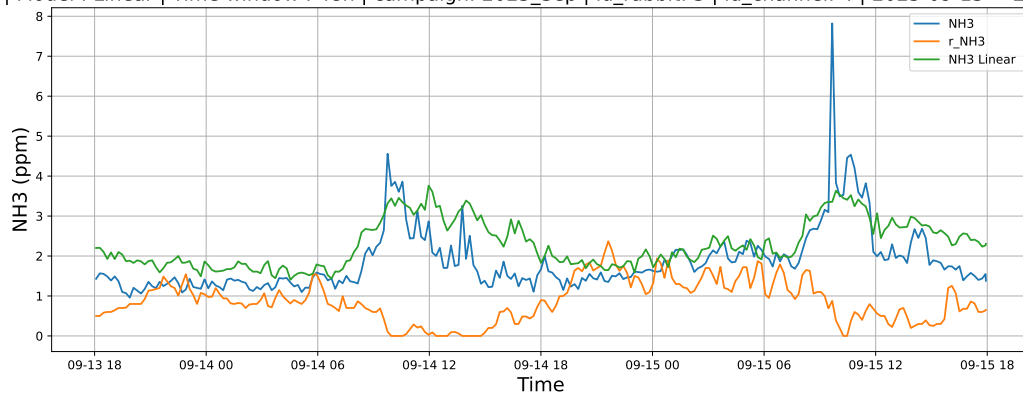


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

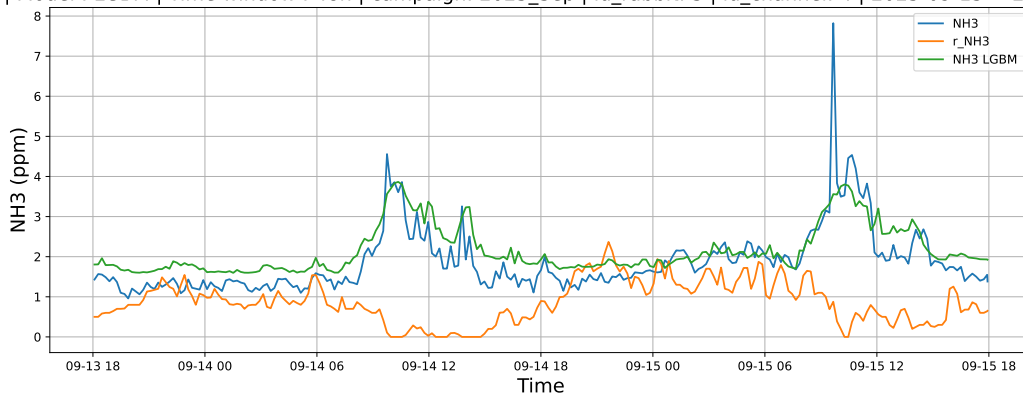


4.1.6.2 Time window : 48

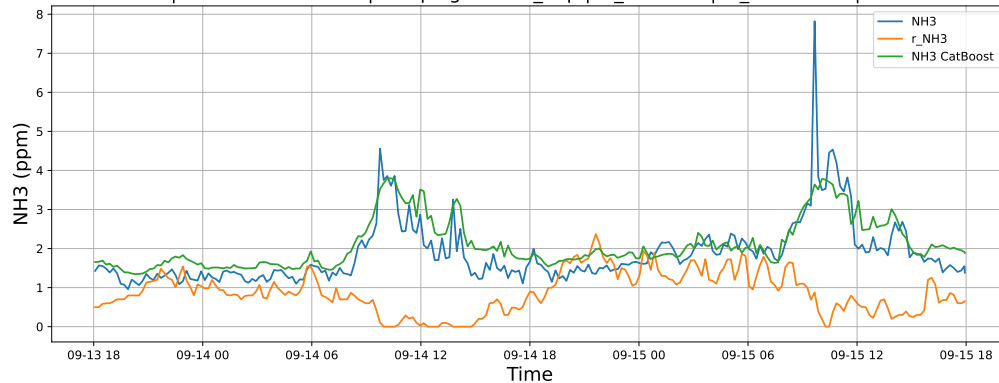
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

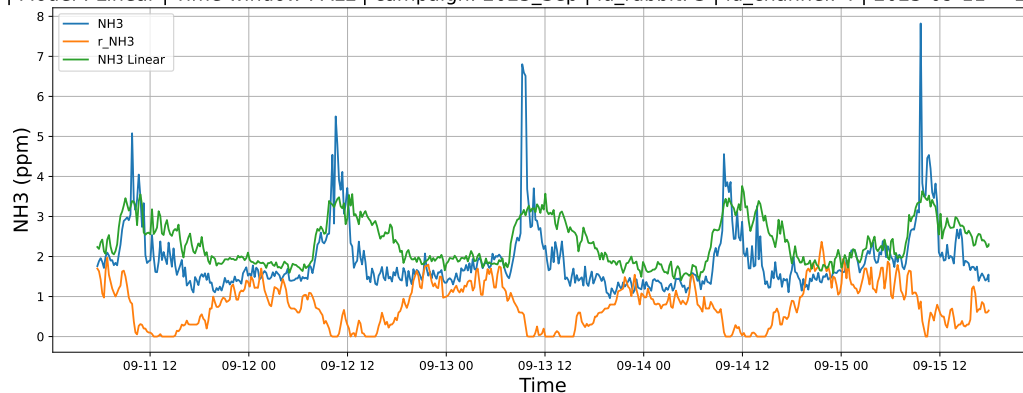


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

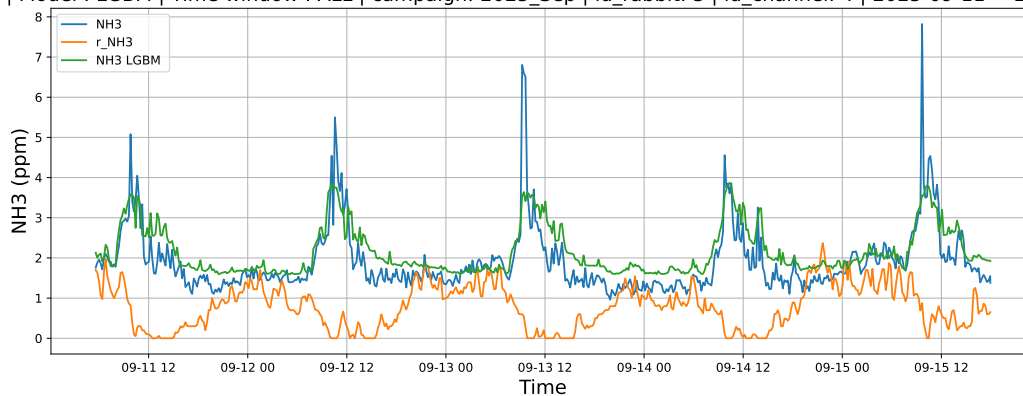


4.1.6.3 Time window : ALL

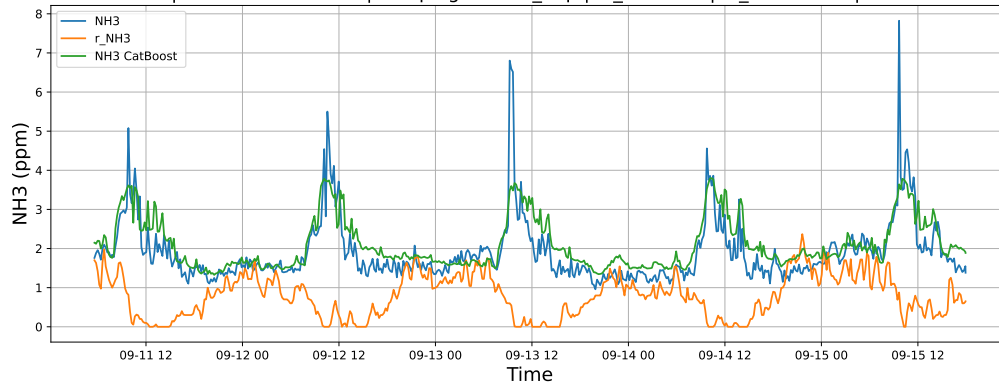
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



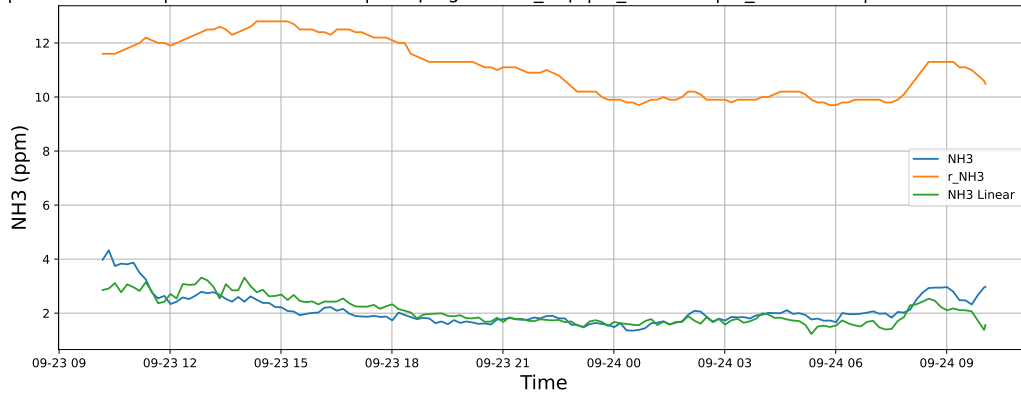
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



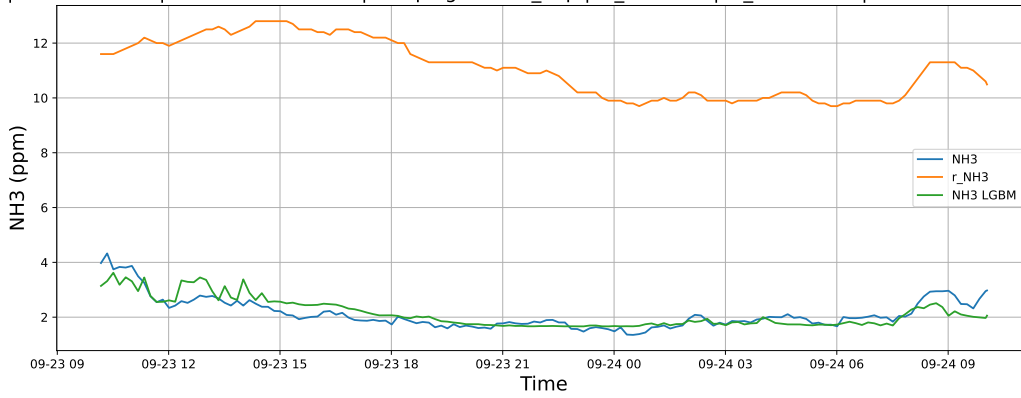
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

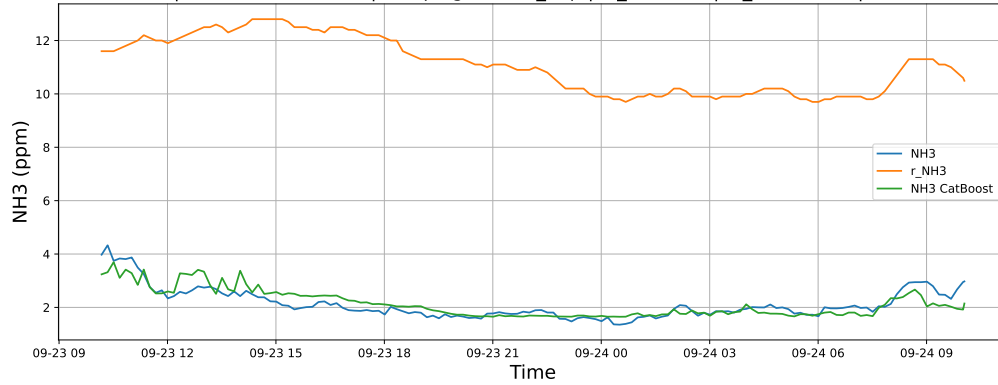
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

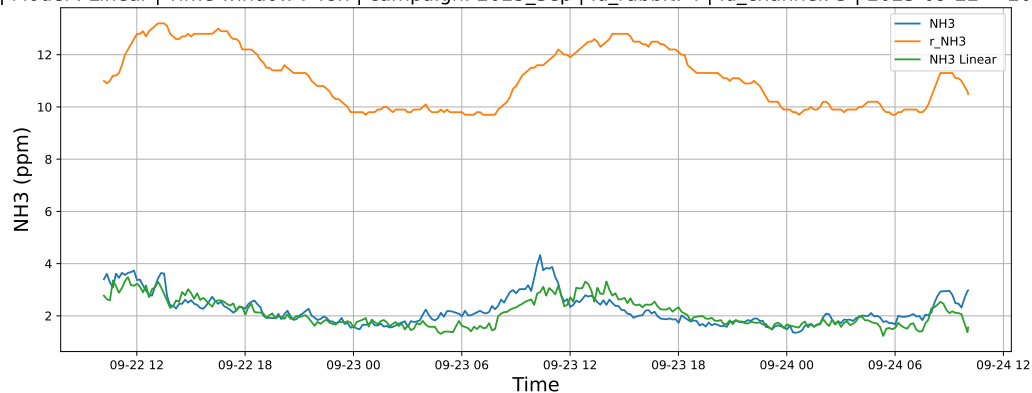


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

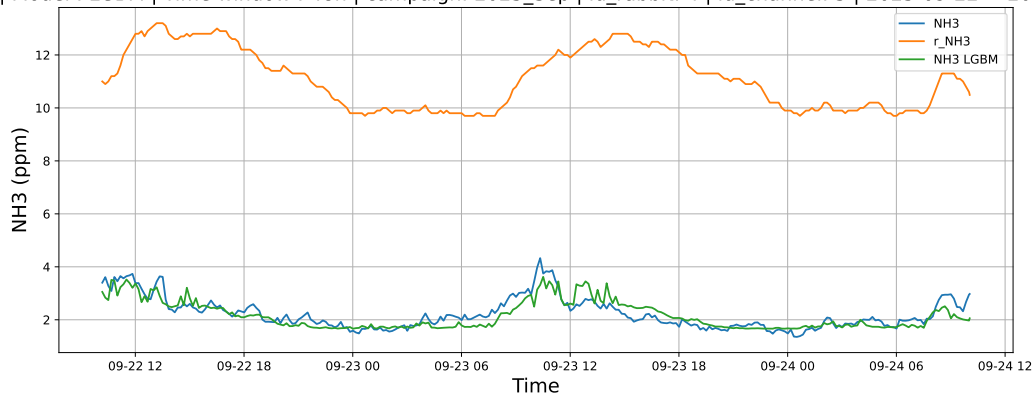


4.1.7.2 Time window : 48

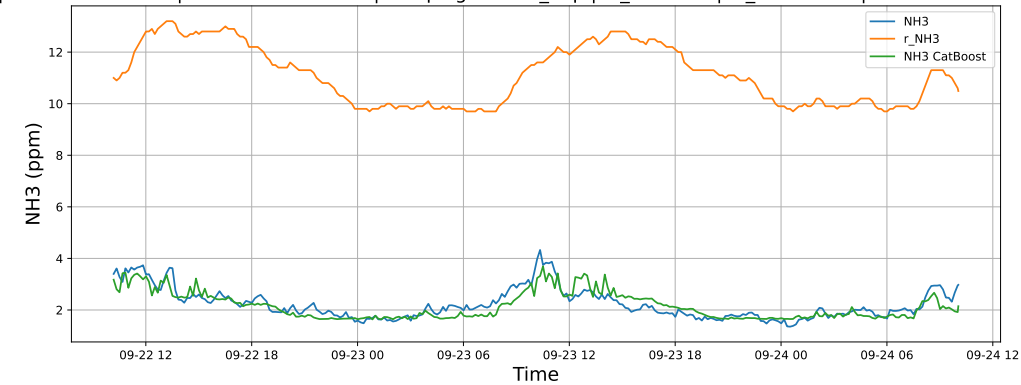
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

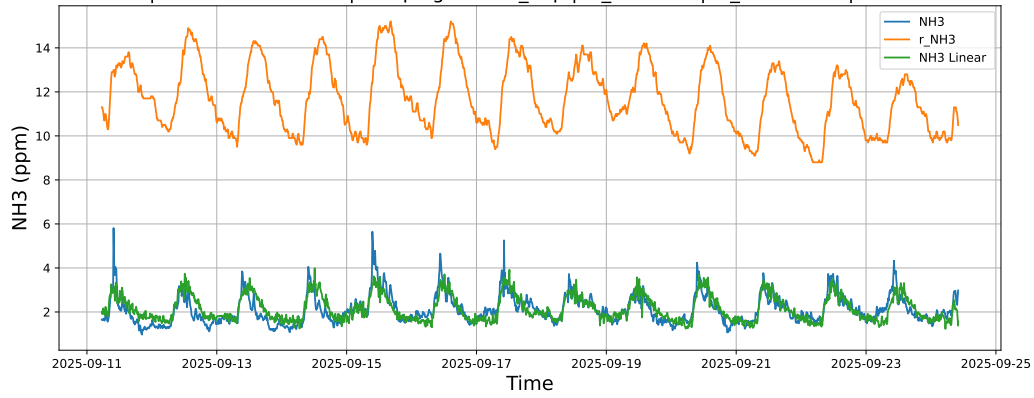


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

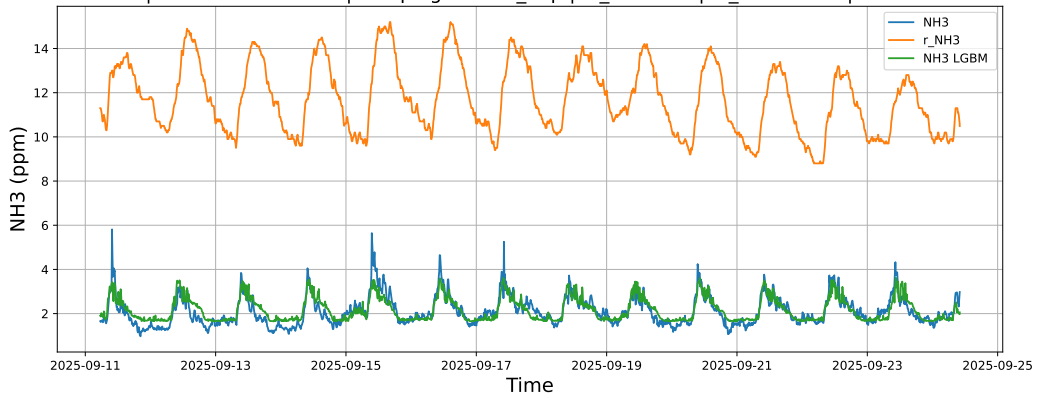


4.1.7.3 Time window : ALL

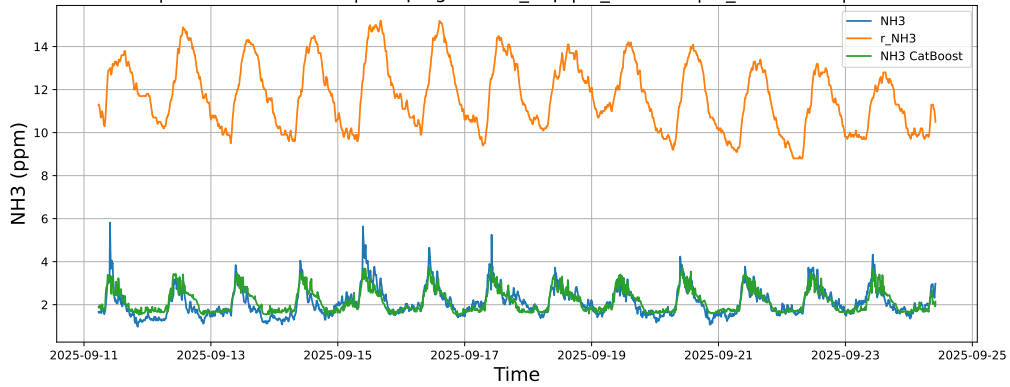
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



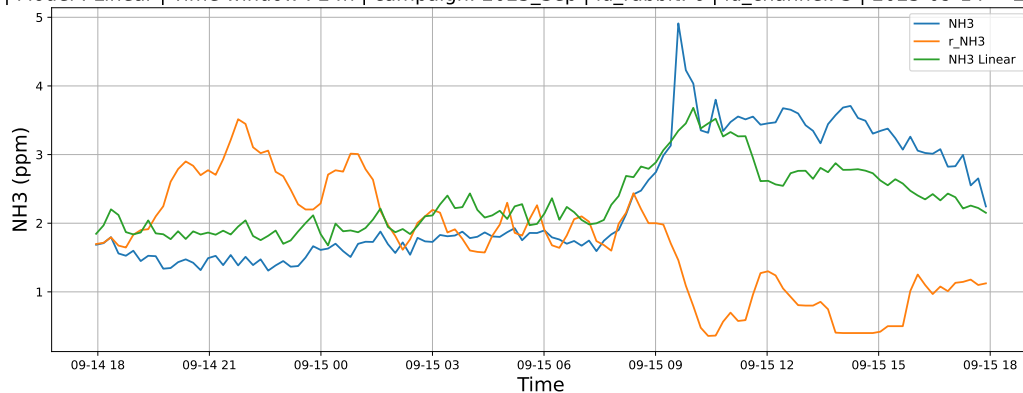
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



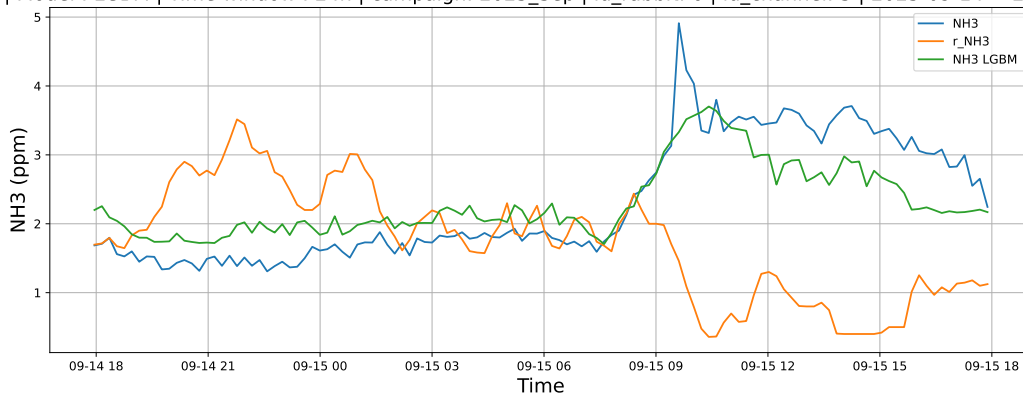
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

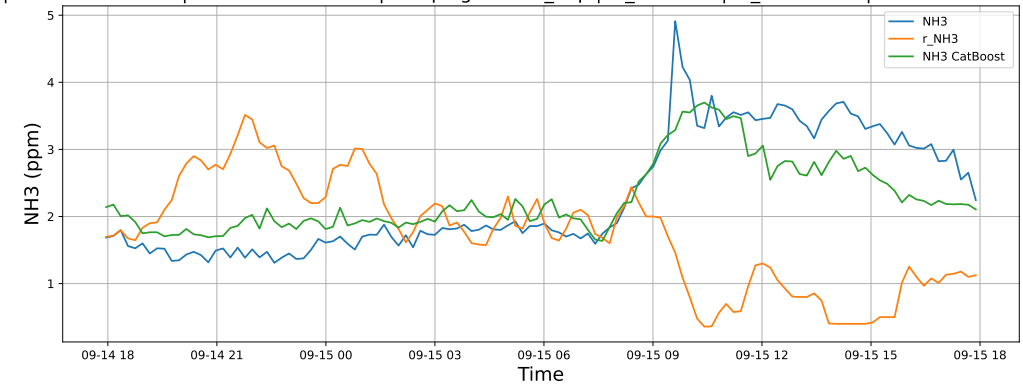
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

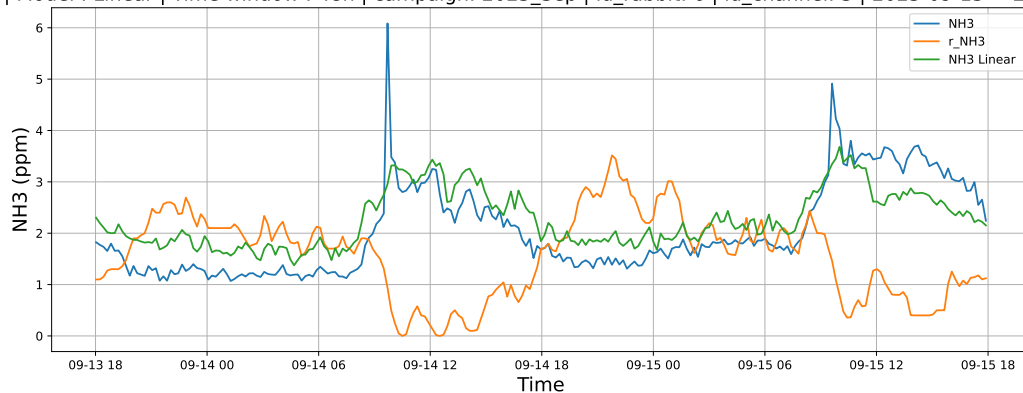


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

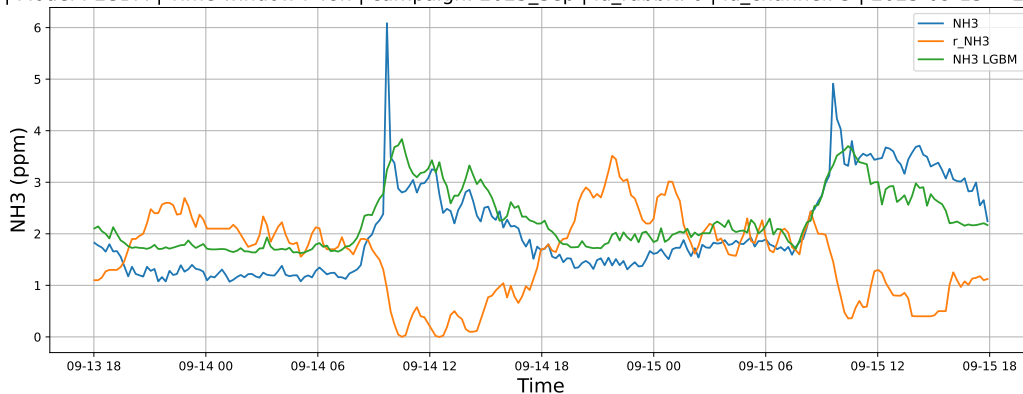


4.1.8.2 Time window : 48

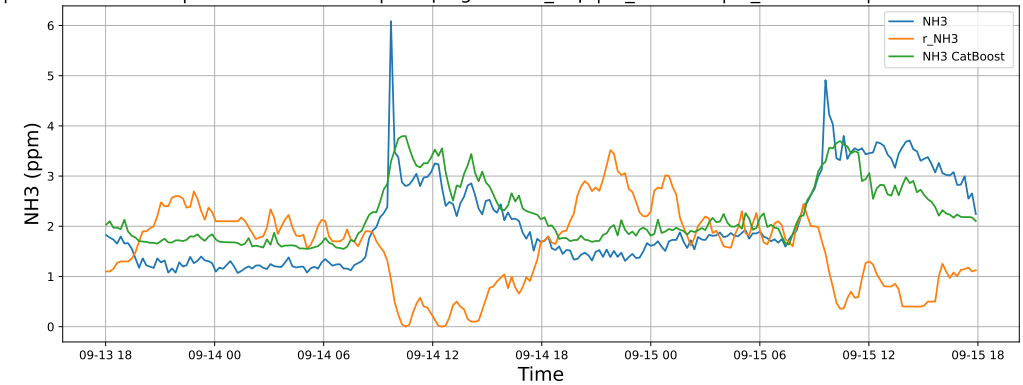
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

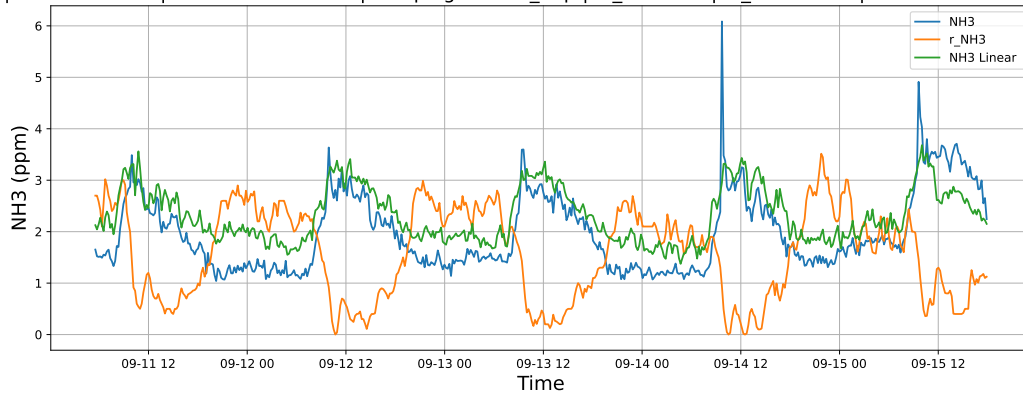


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

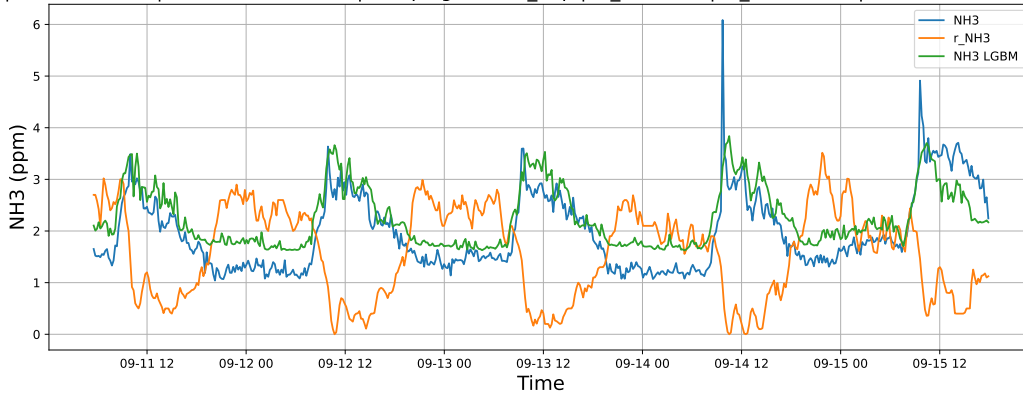


4.1.8.3 Time window : ALL

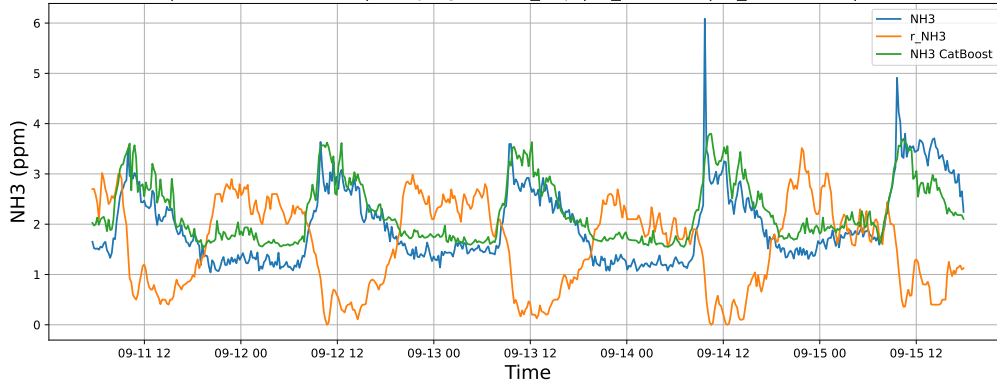
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

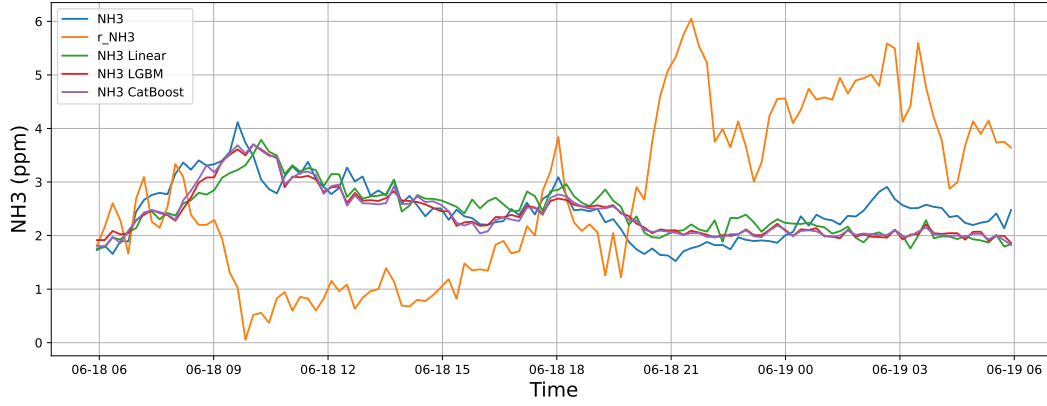


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

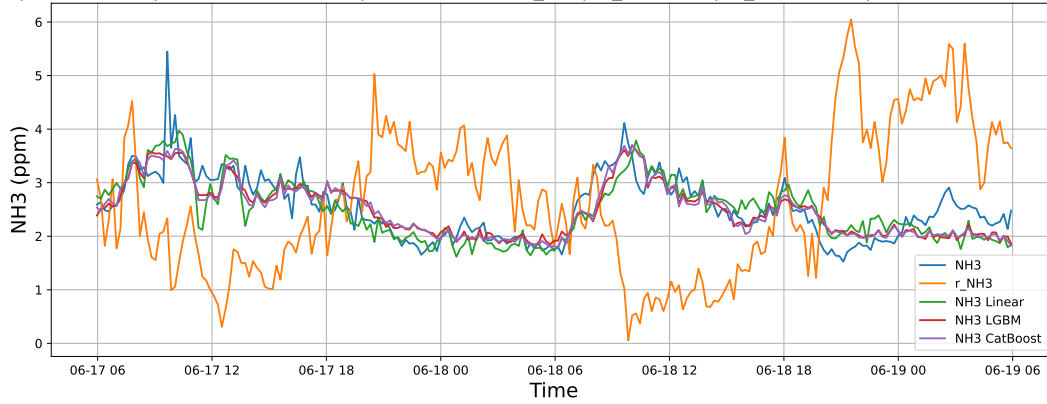
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



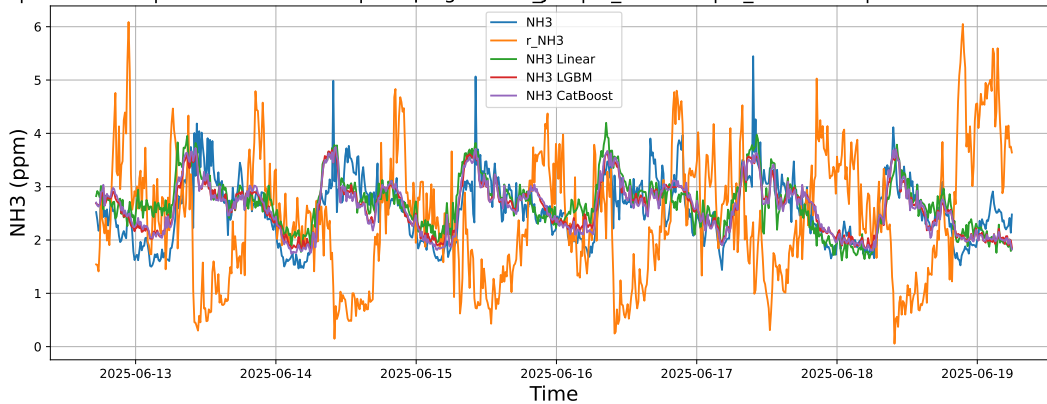
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

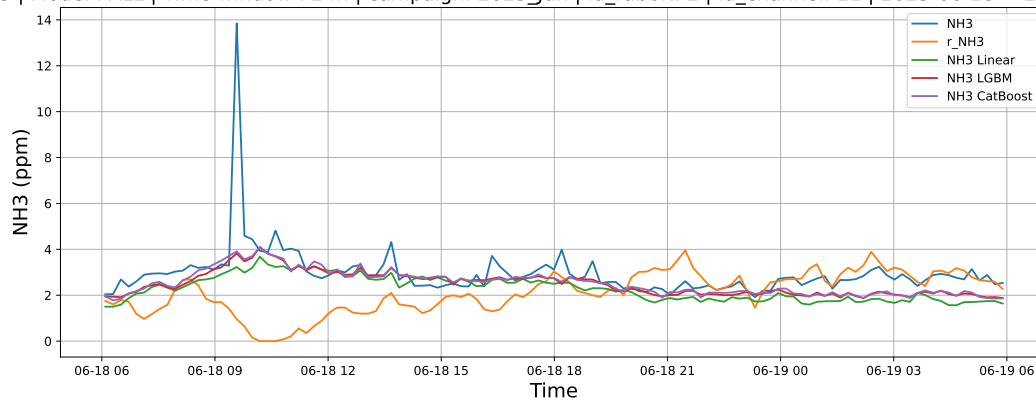
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

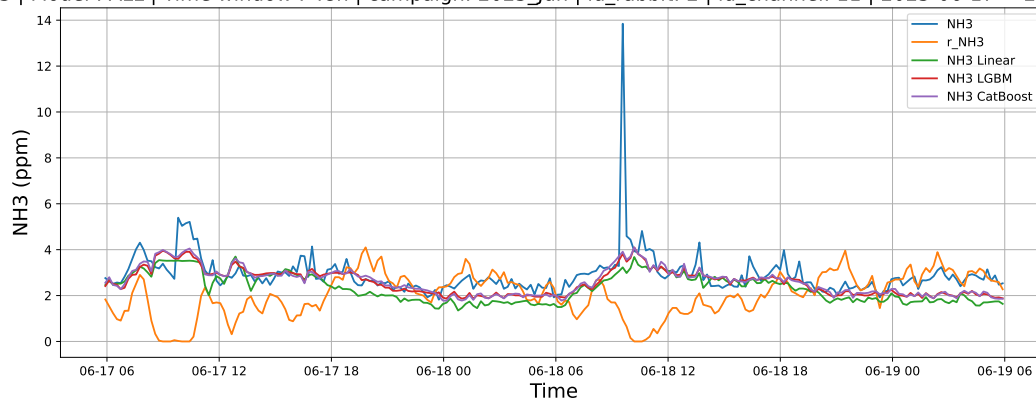
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



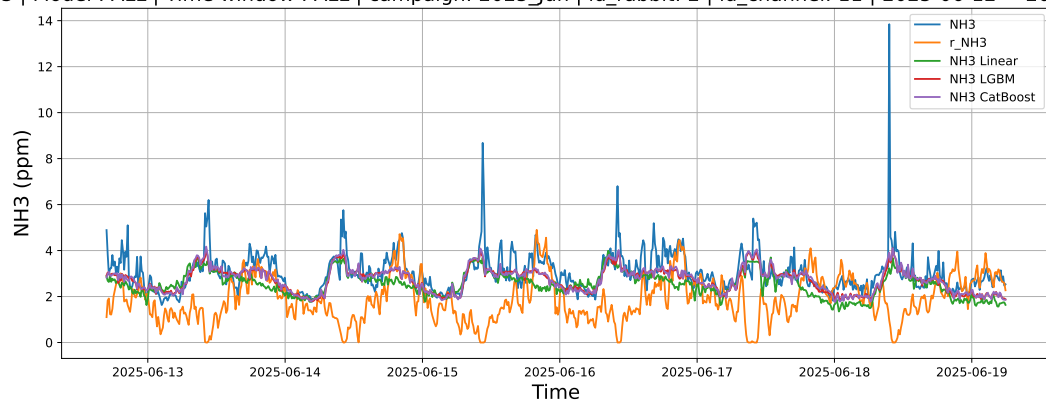
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

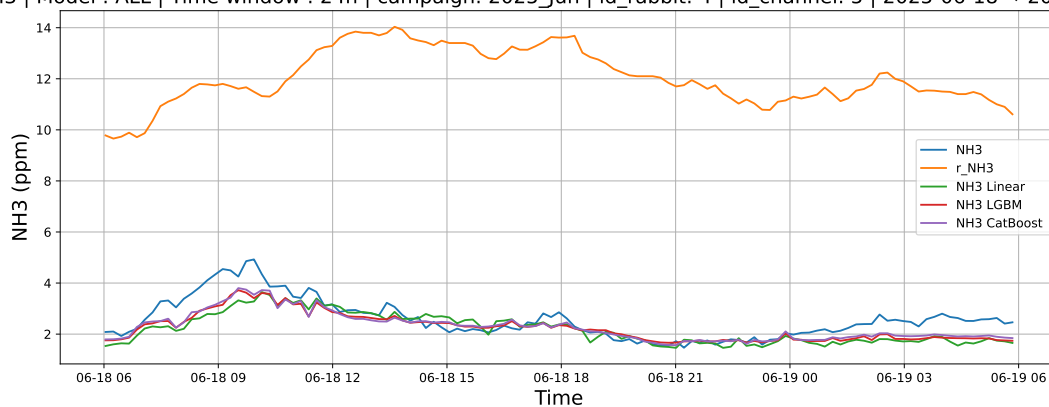
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

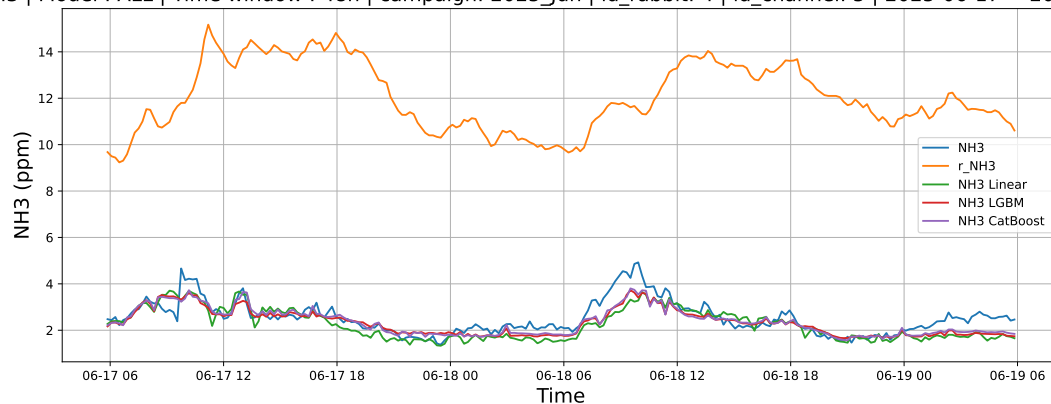
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



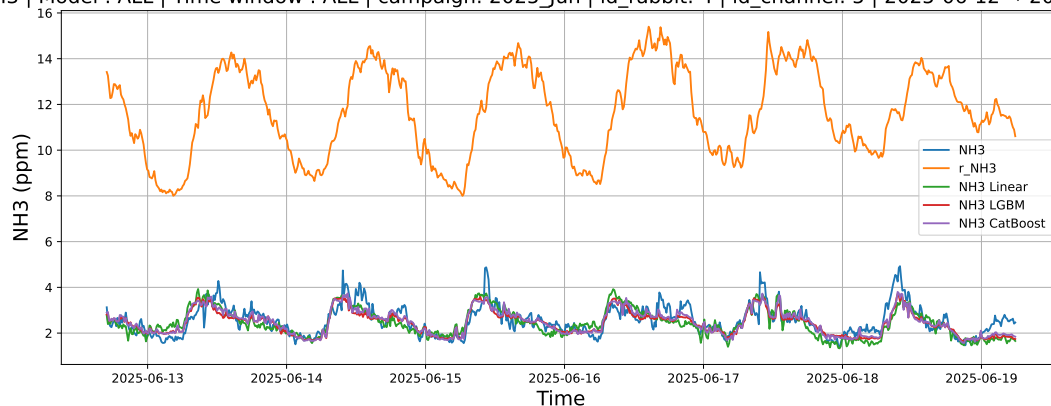
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

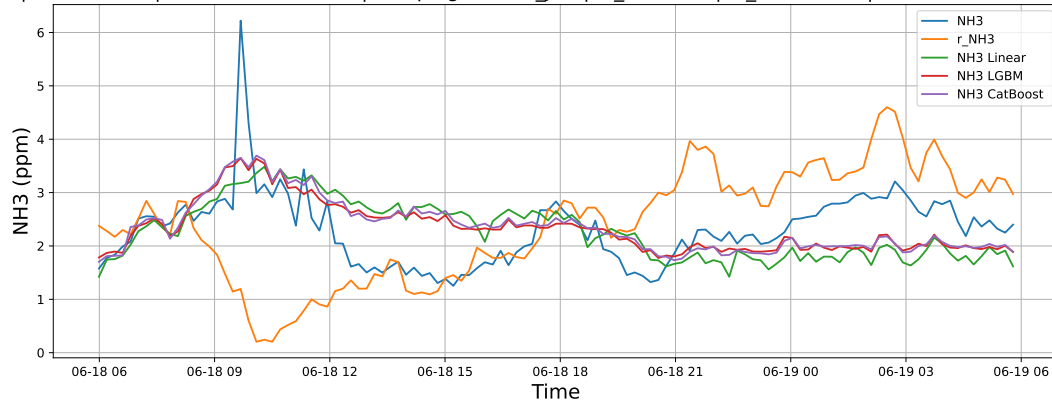
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

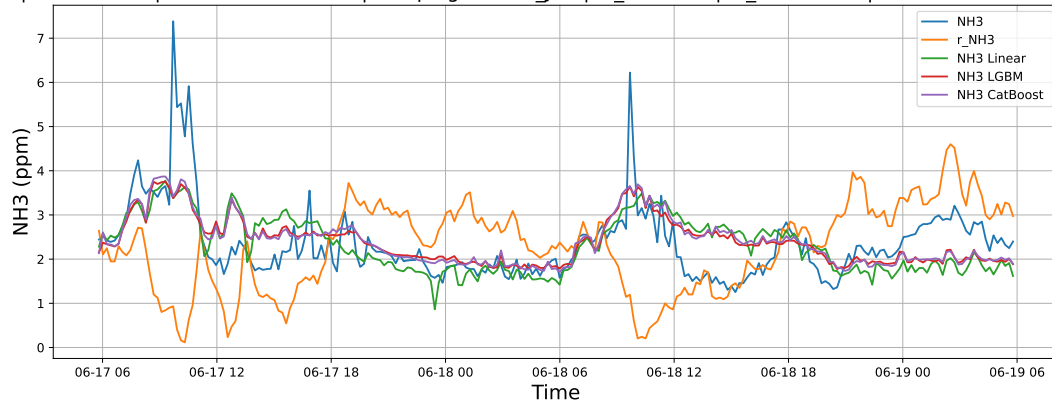
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



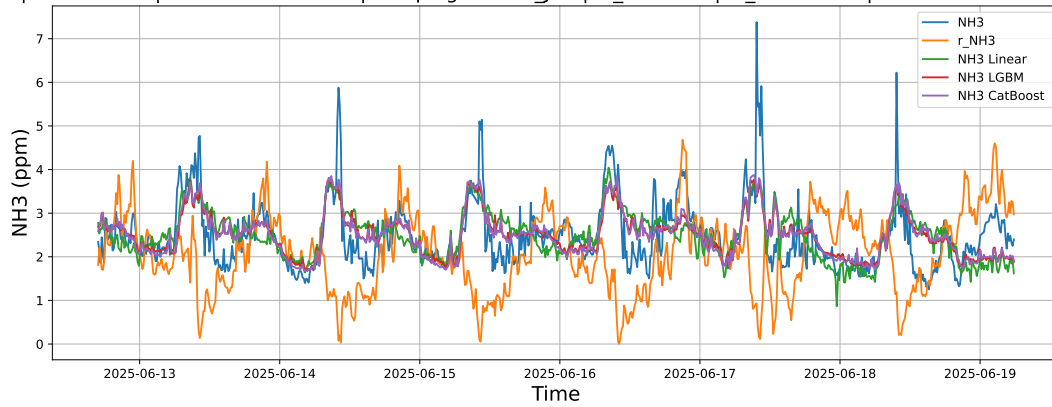
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

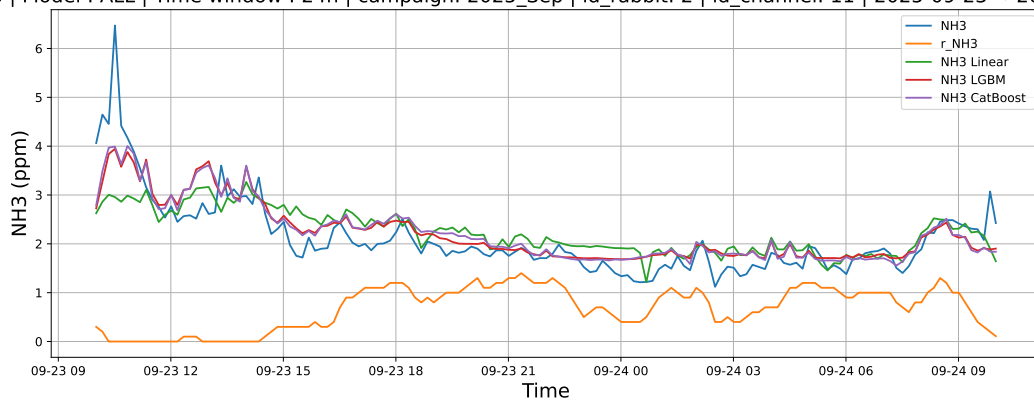
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

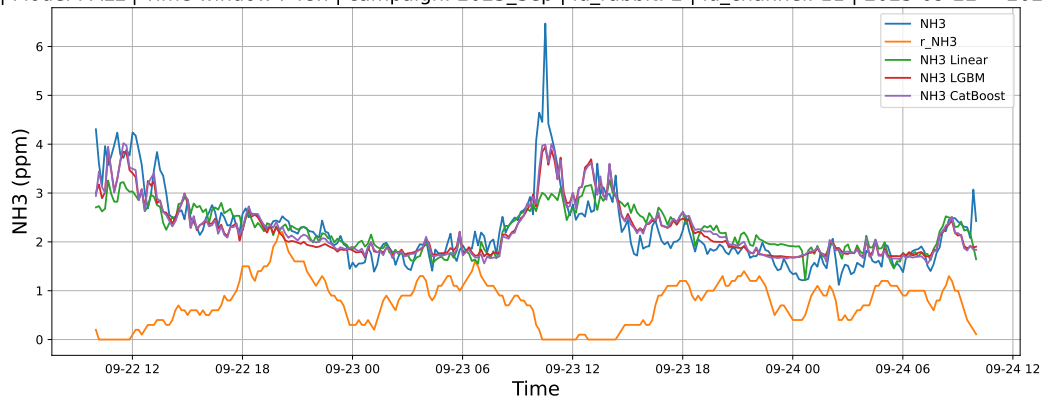
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



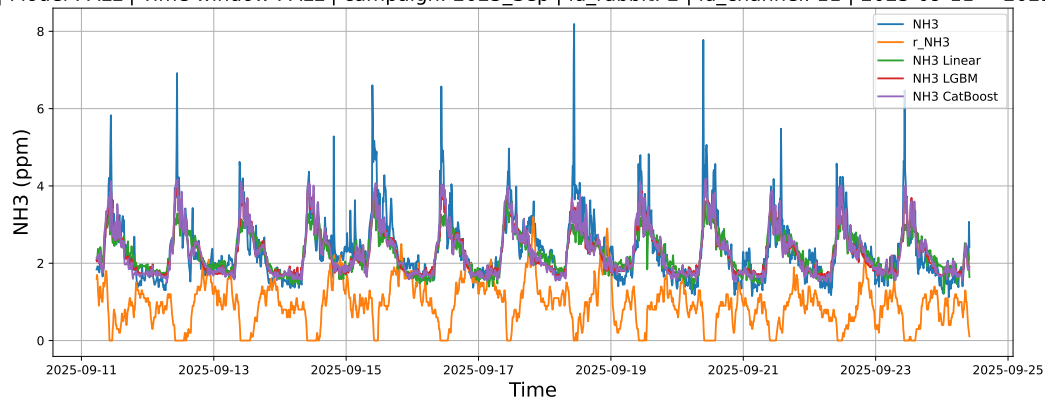
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

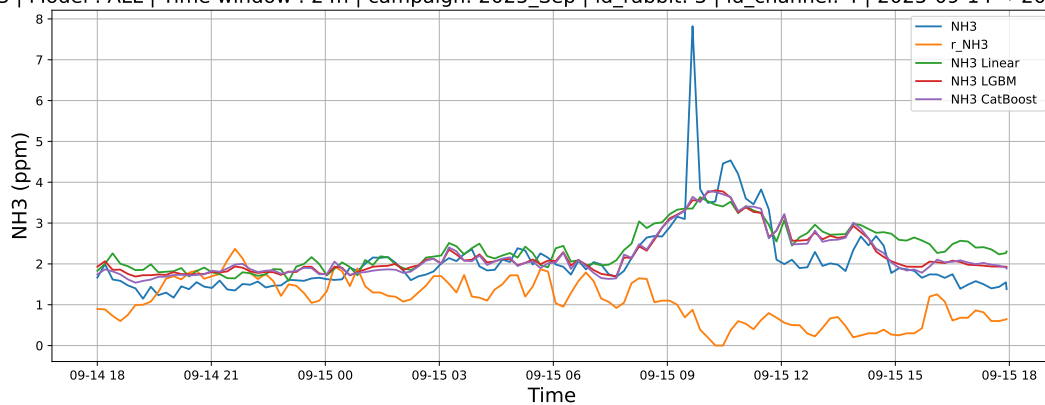
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

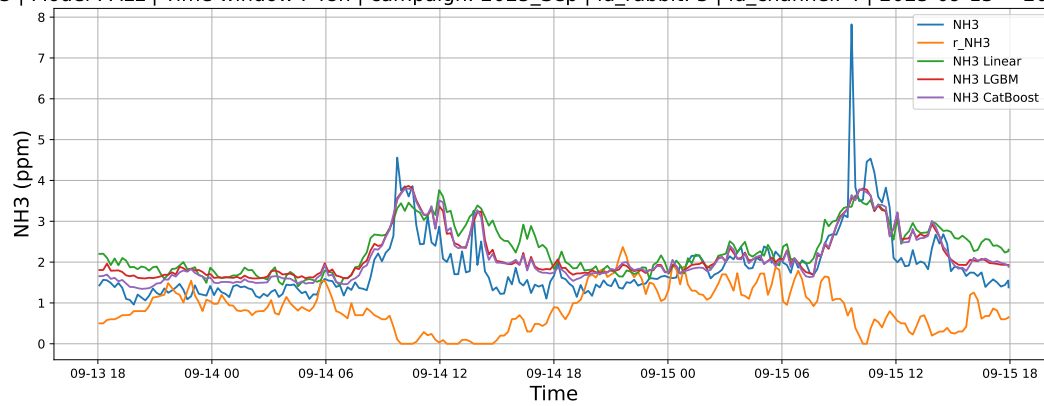
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



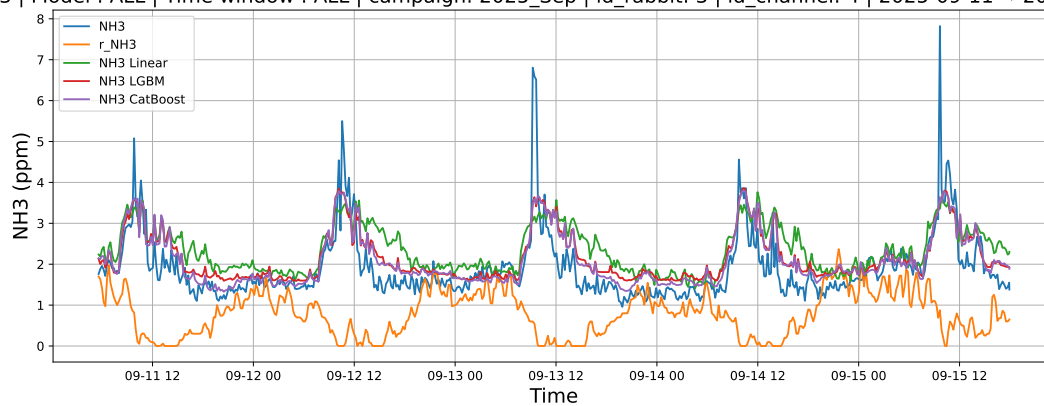
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

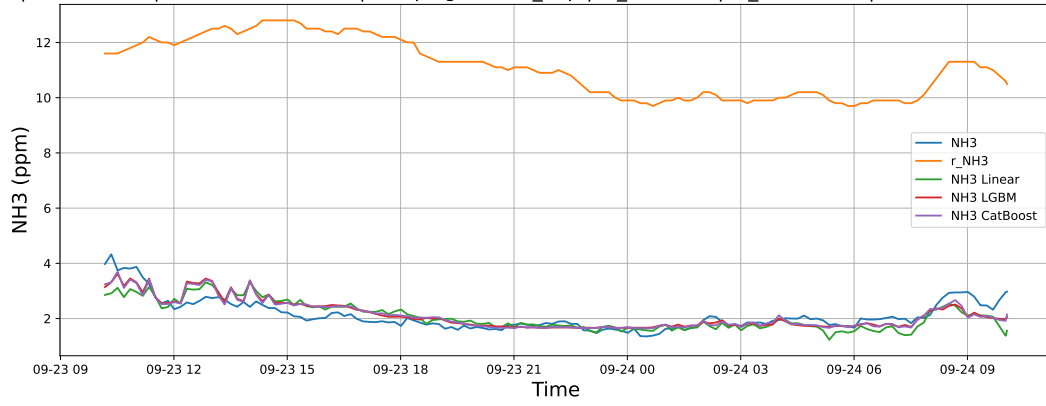
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

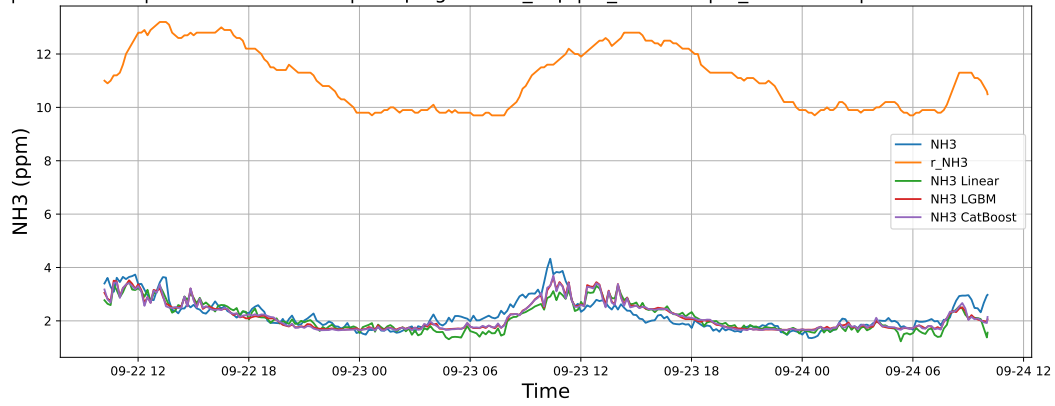
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



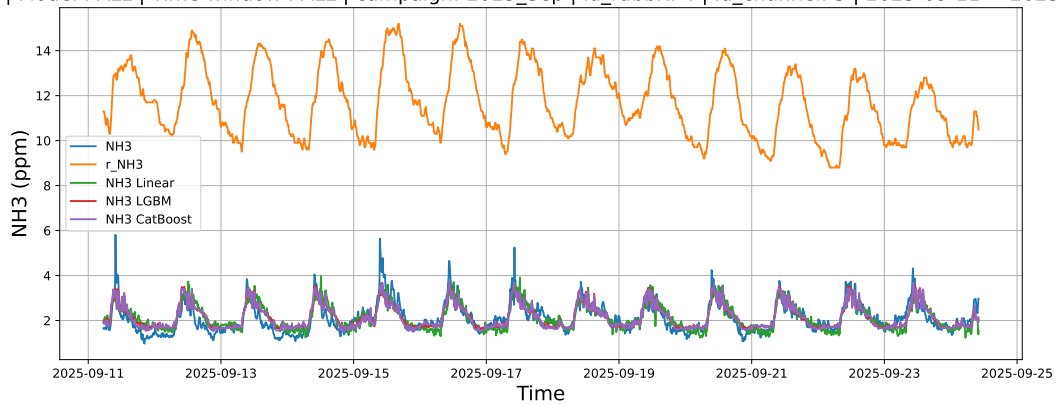
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

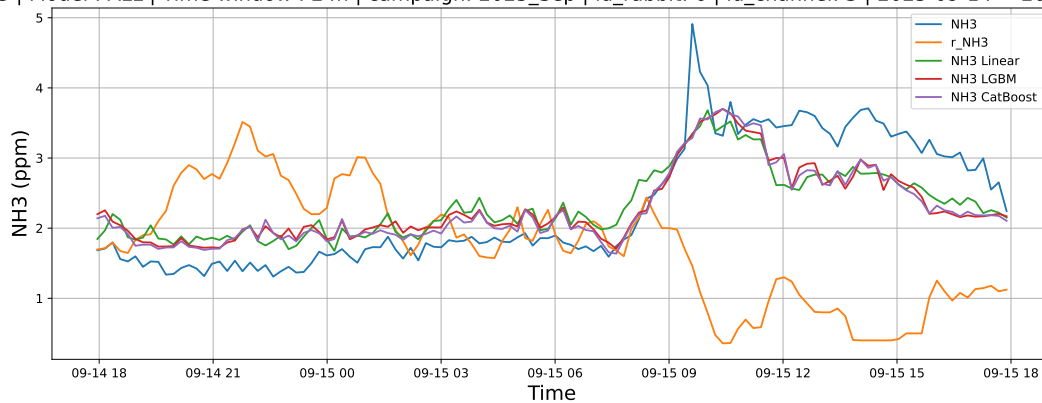
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

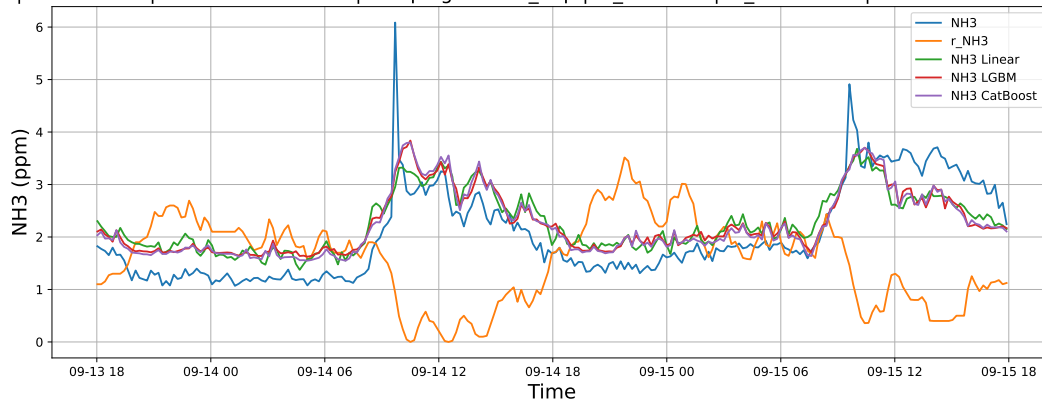
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

